

Droneye – A Quadcopter-Mounted Reflective Transformation Imaging Scanner for Non-Destructive Testing of Wind Turbine Blade Surface Damage

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ABSTRACT

This project set out to design a prototype of a novel drone-mounted Reflective Transformation Imaging (RTI) scanner for the purpose of revealing surface damage in wind turbine blades. It combines the two distinct fields of quadcopter configuration Unmanned Aerial Vehicles (UAVs) and RTI scanners to develop an integrated imaging device. A unique rotating dual-arm mechanism was designed to produce a Multi-Light Image Collection consisting of 96 images, processed using the Cultural Heritage Imaging Foundation's open-source software, Relight, employing the selected RBF+PCA mathematical model. The system was calibrated using stationary test results and mounted onto a custom-built quadcopter, which implemented selected sensors to achieve and maintain a stable position at a fixed distance from a target surface for scanning, as determined through testing. The developed system yielded promising results during static tests across a variety of surface textures, successfully revealing variations in surface depth hidden in standard photographs. However, instability observed in the drone's position-holding function, caused by vibrations induced by uneven mass distribution and limitations in the optical flow tracking system, prevented the prototype from performing as intended during flight. To address these challenges, improvements were proposed, including enhanced sensor fusion for more precise positional tracking, structural modifications to reduce vibrations, and adjustments to the RTI arm mechanism to ensure consistent lighting positions, refining both the scanning system and flight stabilisation mechanisms. This prototype serves as an initial proof of concept, demonstrating the potential of using drone-mounted RTI scanners to assess early-stage wind turbine blade damage, with further development. Future work should focus on integrating advanced inside-out tracking solutions such as Simultaneous Localization and Mapping (SLAM) to improve the device's stability and refining the RTI scanner's lighting control and processing algorithms to enhance scan accuracy under ideal stable conditions in flight, followed by evaluation through tests on real wind turbine blades.

Keywords:

Drone

Reflective Transformation Imaging (RTI)

Wind turbine blade

1.INTRODUCTION

1.1 BACKGROUND

Wind Turbine blades are large-scale composite structures commonly made from carbon or glass fibre reinforced polymers, due to the material's high stiffness and strength-to-weight performance ratio. Varying external factors will affect the blades throughout their lifetimes, including cyclic loading, environmental conditions, and impact stresses over time. These conditions frequently lead to surface-level cracks and defects over the blade's lifetime, including micro-cracks, erosion, and delamination, especially along the leading edge and tip of the blade. With the increase in turbine blade size and length, due to a rise in demand of renewable wind energy, effective inspection has become increasingly difficult due to the challenging, remote environments where the turbines are located. Resultingly, the chosen scanning system must be capable of identifying these types of cracks, whilst being able to be integrated within the drone system, allowing efficient and effective scanning.

1.2 PROBLEM STATEMENT

This project sets out to develop a novel technique for the imaging of wind turbine blades, combining separate fields of quadcopter configuration unmanned aerial vehicles (UAVs) and Reflective Transformation Imaging.

1.3 PROJECT OBJECTIVES

Primary objectives of this project are:

- Construction of an innovative RTI scanner that expands upon existing examples in literature, fitted to the application
- Demonstrate the efficacy of this scanner in revealing surface defects otherwise invisible in simple photography and to the naked eye
- Assembly of a custom drone setup to support the developed RTI scanner and provide stable, viable conditions for scans
- Develop and prove the effectiveness of an automatic process for precisely positioning the drone relative to a target surface

1.3 PROJECT SCOPE

This project will focus exclusively on the use of Reflection Transformation Imaging for surface analysis and employ a drone to position the entire scanner and light setup, differentiating itself from

previous integrations of drone technology and photogrammetric techniques. A simplified case of scanning a surface orthogonal to the device is explored, neglecting the complex three-dimensional geometry encountered in installed wind turbine blades. Furthermore, testing is planned to be conducted at low altitudes to protect the device, avoiding potentially high-speed turbulent wind conditions expected adjacent to fixed turbine blades.

1.5 PROJECT SIGNIFICANCE

This project develops an innovative prototype, with no directly comparable concepts found in literature. As such, it serves as the first step in fusing the two unique technologies of drones and RTI scanners and has been undertaken to assess the feasibility of the concept, providing guidance to future developments of this integration. In the long term, significant development could give rise to a better method of identifying early-stage issues in wind turbine blades' structural health than current, available techniques.

1.6 STRUCTURE AND METHODOLOGY

Firstly, a review of the relevant literature will be conducted. This will provide insight into the background of programmable drones and their application in photogrammetry. The development and significance of surface damage on our intended application of wind turbine blades will be explored, alongside the relevant existing techniques to investigate these, addressing existing uses of drones in this application. The development and workings of Reflective Transformation Imaging (RTI) scanners will then be explained, from the fundamental optical theory to the most accurate processing techniques presently available.

Secondly, the design process of the custom-built drone will be described, outlining the overall scanning procedure and the overall electronics architecture implemented to enable flight. Component selection will then be explained and justified, detailing the integration of a breath of sensors used to accurately determine and monitor the drone's position to create the necessary stability required for scanning using RTI techniques. Critical theory is provided, and design choices for this element of the prototype, which is crucial for overall functionality in the intended application, are built upon extensive research.

Thirdly, the development of a novel, mechanical dual-arm light array is explored, from the initial stages of understanding the requirements of a conventional RTI scanner and the implications of these on a drone-mounted system to developing and refining proposed concepts to form the final scanner design. This includes the Arduino-powered electronics and accompanying code devised to re-light test subjects in an array derived from understanding built upon existing literature, and selection of the optimal mathematical model within available specific RTI software to process results – the culmination of which are integrated with the developed quadcopter drone, refining and tailoring two distinct technologies to our use case.

Once developed, the assembled system was tested in performance of its two major constituent parts - the RTI scanner and the quadcopter drone. Preliminary testing of the scanner was conducted under static conditions to isolate and ascertain the impact of separation distance between the scanner and the subject, the duration of the LEDs' light flashes and the presence of ambient light on the quality of scan results. The optimal parameters determined by these results are later used to calibrate the drone for full-scale testing and are also applied to a series of final stationary tests, yielding ideal RTI scans of a variety of surface types to simulate expected variations in the conditions observed on wind turbine blade surfaces.

The drone subsequently underwent two modes of testing. Both in a loaded and unloaded state. Stationary flight was analysed to assess the performance of the drone's sensor assembly and control systems. A secondary test was carried out to assess the performance of the drones autonomous scripting capabilities, consistently orienting it with a scanning surface along the x-axes. All the aforementioned results are then discussed, providing comparison between critical data to evaluate performance of the developed systems within the prototype. Existing shortfalls in performance are indicated and explained and related to the limitations identified in the drone and RTI scanner portions of the prototype, in all aspects of the incorporated hardware and software. This extends to faults creditable to both the construction processes and testing approaches. From this, suitable improvements are proposed to address all

underperforming elements, encompassing the underlying prototype design, fabrication processes and operation, and opportunities for further development for practical application in industry are described.

Finally, the project's entire course is evaluated, highlighting key conclusions and challenges to assess the overall impact while acknowledging opportunities for future work within the field.

2.LITERATURE REVIEW

2.1 DRONES

2.1.1 Background on Programable Drones

Unmanned Aerial Vehicles, also called drones, were traditionally developed for military applications in the form of either a quadcopter, fixed-wing aircraft or helicopter. As the technology has improved, the underlying principles of drones has remained the same which consists of three main systems: 1. external sensors such as inertial measurement units, barometers, compasses etc. that have some awareness of the drone's position and orientation to provide feedback; 2. a microcontroller that governs the control system of the drone as well as providing auxiliary functions, allowing for flight with varying degrees of autonomy; and 3. a motor driver and power management system that receives instructions from the microcontroller to vary power output to a series of motors as needed.

Initially, these systems were prohibitively expensive and early consumer DIY drones (circa 2008-2012) had to use unspecialised components such as taking the accelerometer and gyroscope sensors from a Nintendo Wii and using an off-the-shelf Arduino as a microprocessor [1]. This changed in 2013 with the release of the original DJI Phantom, which brought drones to the consumer space as well as popularised the quadcopter design [2]. The Phantom also popularised the use of bespoke flight controllers in the consumer space – a board that would combine a microcontroller and crucial sensors (a barometer for altitude and a 6-axis IMU) on a single board and miniaturise it as much as possible, leading to a drone with greater size-efficiency and power-efficiency. Finally, the use of an external GPS allowed for features such as returning to the landing zone and rudimentary stationary flight, implanting the idea of semi-autonomous function.

The Phantom’s innovations of using a quadcopter design to carry a payload, integrated specialised control circuits for flight and use of external sensors for autonomous function, had clear scientific applications. Due to its popularity, this led to the development of the open-source drone solutions that exist today, such as compact flight controllers that use highly programmable STM32 microcontrollers with extensive IO for a wide array of sensors. Additionally, the Phantom led to the adoption of U-blox GNSS (Global Navigation Satellite Systems) modules that are well documented and compatible with the open-source firmwares that were developed to take advantage of the STM32 and similar microcontrollers programmability, namely Ardupilot which was initially built on the Arduino platform and allows for an extensive variety of hardware.

2.1.2 Drone Photogrammetry

Photogrammetry is defined by the (ASPRS) as “the science of obtaining reliable information about physical objects and the environment, through processes of recording, measuring, and interpreting imagery” [3].

The relationship between photogrammetry and unmanned aviation stretches as far back as photography and aviation themselves. Early cameras were strapped to unmanned balloons for aerial photography as far back as 1858 to take photos of city skylines [4]. Eventually this led to the development of the practice of photomosaics, where multiple aerial photographs from a plane would be stitched into a composite map, commonly used in World War I to map German Trenches [5].

For this paper, modern photogrammetric techniques using quadcopters will be considered. There are two primary applications. The first application is the generation of 3D meshes of large areas of land, such as those outlined by Sebastian Siebert and Jochen Teizer [6], who preplanned GPS waypoints that the drone followed using its onboard microprocessor, taking photos along the way. Subsequently, SfM algorithms are used to generate a 3D mesh. This technique can achieve a resolution of several centimetres over a large area despite the low precision of GPS navigation systems due to the high overlap in the images taken, the high resolution of the large mirrorless camera used and the algorithm’s

ability to recognise similar features. Alternatively, for more precise scanning of smaller objects, Théo Louiset, Anthony Pamart and other authors [7], proposed a lightweight drone tracked using an indoor 17 camera outside-in motion capture system. This allows for highly precise drone positioning, albeit in a controlled environment, achieving near centimetre resolution in scanning smaller archaeological artefacts. However, the size and weight constraints of their drone limited them to using a smaller low resolution Raspberry Pi camera. These prior techniques demonstrate how photogrammetry scans on quadcopter platforms involve trade-offs of drone positional accuracy, camera resolution and software image overlap leniency.

Reflectance Transformation Imaging (RTI) scans are a form of photogrammetry and under ideal conditions can achieve sub centimetre accuracy, which if able to be attached to a drone would supersede current resolution limits, potentially for the application of scanning the blades of a wind turbine. This application, however, imposes limitations on a possible design. Primarily, the range the drone must travel would not allow for the highly accurate outside-in tracking solutions such as those used by Théo Louiset, Anthony Pamart and other authors [8], and would instead require an onboard positional solution similar to that used in larger scale scanners, but with a much greater accuracy. Finally, mountings for the lights used by RTI scanners can be spacious and heavy. As such the drone must have a high thrust to weight ratio, with an emphasis on reducing the mass of the drone hardware as much as possible to allow for the scanner to be incorporated at an acceptable battery life.

In recent years, there have been drastic advancements in the onboard processing abilities of drone microcontrollers. This has led to significantly increased functionality of drone firmwares, allowing for the use and programming of a much wider array of sensors than was typically available. This has occurred alongside drastic miniaturisation in the field of camera sensors, which has allowed the authors to create the following RTI drone prototype concept which prioritises the use of outside-in tracking to supplement traditional GPS positioning, enabling high precision tracking and semi-autonomous function on a lightweight platform.

2.2 NON-DESTRUCTIVE TESTING

2.2.1 Cracks and Defects

Structural defects in wind turbines represent a critical threat to safety, operating efficiency, and the longevity of the structure. Due to the extreme forces present when operating wind turbines, extreme environmental conditions, and manufacturing and production errors, a variety of cracks and defects can occur through the blade's lifecycle. These defects include surface damage, delamination, structural cracks, leading edge erosion, voids, peeling and matrix cracks.

Research into the improved efficiency of drone utilisation in structural health monitoring (SHM) technologies makes it clear that "Understanding the nature of damage in wind turbine is essential for optimising inspection technologies and preventing catastrophic failures" [9]. Early detection and the correct classification of defects is essential for the development and integration of SHM systems. This section aims to understand the different defects that can impact wind turbine blades, using recent studies to illustrate their implications.

Figure 1 shows the types of surface damage defects that can occur on the blades, which include microcracks, erosion, and coating degradation, usually caused by environmental conditions. These defects can act across the whole surface of the blade, but are mainly concentrated along the leading edge, and tip regions where the blade is subject to erosion from "rain, hail, and other particles" [10].

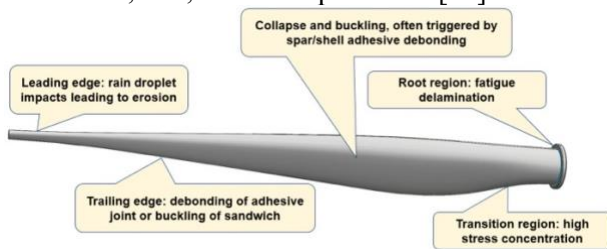


Figure 1: Locations of often-observed damage mechanisms of a wind turbine blade [11].

Mishnaevsky classifies wind turbine blade damage into "surface damage" and "structural element damage" [11]. Surface damage initially starts out as small cracks or defects, but over time will propagate becoming more severe internal failures, if left unchecked.

Delamination can become a severe defect, as it occurs when "different layers of the composite material separate due to fatigue loads or

manufacturing defects." [12] and is usually hard to detect visually using traditional sensing methods. Delamination is the result of manufacturing defects, weak resin bonding or cyclic fatigue. Fayyad states that delamination failure is most common on regions of the blade where "high shear stresses are present, such as near the blade root and spar cap interfaces." [9]. Delamination works with other defects (matrix cracks) and compromises the load-bearing capacity of the blade, effecting its aerodynamic properties and causing an increase in the shear forces present. These effects eventually result in failure, if not detected efficiently.

Structural cracks can be a serious problem to the integrity, stability, and safety of the blade if left unaddressed. These cracks are initiated by fatigue loading and are "commonly located at areas of high stress concentration," [11]. These cracks occur internally within the structure of the blades' trailing edge, spar cap, and ply drop-offs. The cracks may appear as transverse cracks, depending on the orientation of the load on the blade. Structural cracks are a result of fibre misalignment, defined using the equation: $\sigma_{failure} = \sigma_0 \cos(\theta)$ [13]

Where $\sigma_{failure}$ is the failure stress, θ is the angle of fibre misalignment and σ_0 is reference strength for the material of the blade.

Leading edge erosion (LEE) refers to the gradual erosion of a turbine blades' leading edge, caused by environmental factors such as rain, hail, and sand particles over a prolonged period. LEE is one of the most common surface degradation mechanisms within operating turbine blades. The constant erosion of the leading edge "causes degradation of the blade material and influences aerodynamic properties" [10], altering the efficiency of the blade, and stripping the blade of its protective coating, furthering its deterioration.

Some defects can occur from manufacturing. For example, voids act as crack initiation sites within the blade, whereas matrix cracking degrade the surface integrity of the blade. Voids are empty pockets of air that form inside the moulded parts and "occur due to improper resin flow during manufacturing" [14]. Matrix cracking results from cyclic loading, resulting in the formation of micro-cracks within the matrix material of the blade. The internal resin that holds the material together starts to break down, causing failure.

Crack detection through image analysis will be used through the project, utilising the equation to aid in

the visual measurements of these physical cracks:
 $L = NP$ [14]

Where L is the physical crack length, N is the number of pixels along the crack, and P is the real-world length per pixel.

Each of these cracks and defects, whether they are external or internal, can lead to complete failure of the blade. Therefore, early crack and defect detection is essential for the prolonged lifetime and cost-effective maintenance of the blade. The RTI system aims to detect early signs of the defects, previously mentioned, before they become dangerous.

Unplanned blade repairs are a huge problem in the wind and renewable energy industry, as “onshore wind farm operators spent around USD 15 billion on operations and maintenance services in 2019, 57% of which was on unplanned repair” [11], highlighting the importance of early detection and action, as blade defects and failures can also result in lower power generation.

Wind turbine blades “have a designed lifetime of 20 to 25 years” [15]. However, undetected defects, internal structural issues, can drastically reduce their expected operational lifetimes. By being able to detect and categorise these defects earlier and more efficiently, the RTI system has the potential to extend blade lifetimes, reduce maintenance costs, and reduce harm to potential inspectors.

2.2.2 Current Testing Methods

Wind turbine blades need to be efficiently inspected using non-destructive evaluation (NDE) methods identifying damaged caused in installation and in-service. This avoids catastrophic failure and maintains suitable aerodynamic conditions for improved efficiency. There are several different techniques used in the inspection of wind turbine blades, which can be divided into two groups: mechanical wave-based methods and electromagnetic methods.

Ultrasonic testing (UT) (mechanical wave-based method) is commonly used in the detection of delamination-type internal defects in fibre-reinforced blades. UT propagates high-frequency sound waves through the laminate [16]. Pulse-echo or through-transmission UT can disclose delamination or disbands and even provide C-scan maps of subsurface damage. For instance, UT has been employed to estimate the level of adhesion between individual layers [17]. However, there is a

limit on the resolution of the C-scan map due to the need for lower frequencies to penetrate thicker sections as the heterogeneous multilayer structure of the composites can cause strong wave scattering [16]. This method can also be time consuming and requiring direct human access to the blade itself by either climbing or removing the blade.

Another mechanical wave-based method is acoustic emission (AE), which involves detecting the high-frequency stress waves emitted when a material undergoes damage such as crack growth. AE sensors on the surface of the wind turbine blades will detect the sudden release of energy emerging cracks or delamination and provide the engineer an instant update [17]. This method is useful for detecting fine microcracks in the fibreglass but is prone to false positives and requires skill to interpretate the signals from sensors as noise caused from machinery and wind can cause similar signals to cracks [18]. While this method does not require direct human access, it is limited by the high chance of false positives and a need for a skilled individual to interpret the wave signals.

Infrared thermography is a type of electromagnetic based method that scans broad areas of the blade by creating a temperature pattern at the surface. Internal defects distort the flow of heat and produce surface temperature anomalies, which is captured by IR cameras. Confirmation exists of successful inspection of defects in the blade using passive IR, including trials with IR cameras onboard drones to inspect installed blades remotely [19]. Field trials have demonstrated that drone-deployed infrared thermography can produce clear thermal images of blade defects [20]. Thermography's major advantages are speed and inspection area but it's largely sensitive to near-surface defects compared to deeper defects. However, the wrong conditions can create challenges and cause false positives. This method also is best used for subsurface cracks and defects as opposed to surface defects that can affect the aerodynamics of the wind turbine [21].

Laser shearography is an optical interferometric technique used to measure small gradients in surface deformation, making it highly suitable for detecting subsurface flaws in composite blades under load. In laser shearography, the surface of the blade is stressed (e.g., using gentle vacuum application,

thermal loading, or vibration), and the existence of any internal flaw (e.g., delamination) causes a localised strain anomaly, which appears as fringe patterns in the resulting shearography image. According to Dantec Dynamics, a company specialising in the development, manufacture and application support of measurement systems, states that laser shearography, which usually cannot be found using traditional non-destructive methods, with rates as fast as 2 metres per minute [22]. This is a great advantage for large structures. However, a main limitation is that it requires a vibration-free and stable environment when taking measurements. The blades in the field are subjected to wind and ambient vibrations, which can interfere with speckle interferometry; therefore, traditional shearography has largely been limited to testing in factory or laboratory environments. Efforts are underway to enable shearography in the field, including the use of robotic platforms that are intended to be mounted to the blade surface for the purpose of sensor isolation [23].

Radiographic inspection is a method for internal blade visualisation by exploiting differences in material thickness and density, which reveals voids, cracks and fibre misalignments by detecting changes in x-ray absorption through the composite [17]. High-resolution radiographs can clearly show air voids or porosity in adhesive joints and laminate stack-ups that might be missed by surface inspections [17]. The main limitation of such method is the logistical aspect of using powerful radiation sources and large detectors for full coverage, meaning that the scans must occur offsite. While radiography yields rich information and can accurately pinpoint internal flaws (with low false-negative rates when done properly), its real-world application for in-service blade inspection remains limited by equipment requirements and operational costs [24].

Magnetic Particle Inspection is a method exclusively used for ferromagnetic metals. The component is magnetised, and iron particles are applied to the surface, causing the surface to near-surface cracks to disrupt the magnetic flux and the particles accumulate at the leakage fields and visually reveal the crack. MPI is highly effective for detecting fine cracks in steel structures but is limited as it requires the test material to be ferromagnetic. Most wind

turbine blades are made from fibre-reinforced polymer composites, meaning that MPI cannot be used to inspect the composite blade material [25]. However, MPI may be useful in wind turbines for metallic attachments, fasteners or other parts, but not the blade itself. Due to this reason, as well as the inconvenience and difficulty associated with conducting this without removing the sample from the turbine, MPI is rarely or never used for wind turbine inspection.

Eddy Current Testing uses electromagnetic induction to detect flaws in conductive materials. An energised coil produces an oscillating magnetic field that induces circulating currents (eddy currents) in the test object. If the eddy current flow is disrupted by a crack or other defect, the impedance of the coil changes, indicating the presence of a flaw. ECT is widely used in the aerospace and power-generation industries to find faults in metallic turbine blades [26, 17]. However, like the MPI method, ECT is disadvantageous for composite wind turbine blades. The blade's fibreglass material is an electrical insulator, and even carbon-fibre composite sections have far lower conductivity than metal [27]. As a result, a standard eddy current probe induces only very weak currents in a composite, yielding little useful signal for defect detection.

In recognition of these limitations, researchers have investigated adapted or hybrid electromagnetic techniques for composite blade inspection. One approach is to use higher excitation frequencies and specialised coil designs to improve the eddy current interaction with carbon-fibre materials [28]. Another promising strategy is eddy current thermography, a hybrid method in which eddy currents (often induced by a magnetic coil or microwaves) locally heat the material, and an infrared camera captures thermal contrasts to highlight subsurface defects. This technique has shown the ability to detect impact damage, delamination, and surface-breaking cracks in composite specimens by virtue of the heat generated at those flaw sites [27]. For instance, pulsed eddy current thermography has been successfully used to identify barely visible impact damage in carbon fibre coupons, offering fast scanning and a relatively wide defect detection range [29]. While such innovations are still largely in the research phase, they illustrate how ECT can be augmented to inspect non-metallic materials.

Visual inspection is the most traditional and widely practiced blade inspection method, encompassing any technique where an inspector or camera directly observes the blade's surface. In manual inspections, technicians either climb the turbine blade (via rope access or suspended platforms) or use telescopes and binoculars from the ground to scrutinise for visible defects. Common damage indicators—cracks along the surface, trailing edge or leading-edge splits, gelcoat blistering, lightning strike burn marks, or erosion of the paint can often be identified by a trained eye [17]. However, this process is labour-intensive and slow: a rope-access team typically requires several hours to inspect all sides of one large blade (during which the turbine must remain shut down) [17]. Furthermore, sending personnel up-tower is costly and poses safety hazards, especially in offshore or high-wind scenarios [30]. Even with careful attention, manual visual inspection may miss small or inception-stage defects, and its effectiveness largely depends on the experience of the inspector. These limitations have motivated the wind industry to augment and partially replace manual blade checks with automated visual methods.

2.2.3 Scanner Technology Evaluation

A range of optical scanning technologies were considered for detecting surface defects on wind turbine blades. Commonly used scanners and methods will be evaluated, such as structured light scanners (SLS), laser scanning (LiDAR), and photogrammetry. This evaluation will showcase the rationale for selecting RTI as the most suitable scanning technique to be integrated into UAV technology.

SLS project a known pattern onto a surface and use deformation of that pattern to reconstruct 3D geometry. SLS provides high-resolution surface data and is widely used in precision metrology, highlighted by table 1. This relies on triangulation between the projector and camera to compute depth at each pixel, enabling high accuracy (sub-millimetre). However, SLS performs poorly on shiny, transparent, or highly textured surfaces, as these surfaces can distort the project pattern.

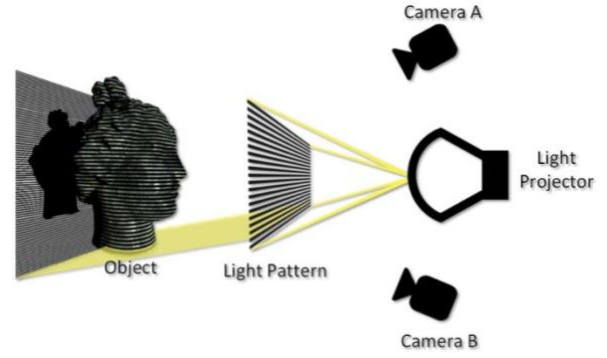


Figure 2: SLS method diagram [31].

Laser scanning, including aerial LiDAR, is highly effective for mapping large-scale structures and capturing geometry under varied lighting conditions. “Laser scanners can generate dense point clouds capable of capturing the fine surface structure of cultural heritage and infrastructure objects” [32], as pointed out by Bretzke. Photogrammetry uses 2D images taken from multiple angles to reconstruct 3D surfaces. While effective for larger objects, its resolution depends heavily on camera quality and lighting control.

Scanner	Accuracy	Size / weight	Power Demand	Drone Suitable	Cost	Lighting Controller
SLS	High	Medium	High	Limited	High	Yes
LIDAR	High	High	High	Poor	High	No
Photogrammetry	Medium	Low	Low	Yes	Low	Yes
RTI	High	Low	Low	Yes	Low	Yes

Table 1: Scanner Comparison Table.

Although SLS and LiDAR offer superior 3D precision, they are typically bulkier, power intensive, and require precise distance control. These factors have a very limiting effect on the integration with a small UAV system (table 3). Photogrammetry is lightweight and flexible but its accuracy “is largely dependent on lighting, texture, and image resolution, and it performs best under controlled photographic conditions” [33], unsuitable for varying environmental conditions.

RTI was selected for this project due to its balance of lightweight setup, low power requirements, and ability to reveal fine surface textures using directional lighting. RTI works by capturing a series of images under varying light angles and computationally modelling how light interacts with

surface features, making it especially effective for detecting shallow surface cracks and defects [34]. RTI also supports a flexible lighting setup, allowing it to be integrated into a physical drone-mounted system. Unlike SLS, RTI does not require a fixed stand-off distance or pattern project, and unlike LiDAR, it imposes minimal demands on onboard power and weight.

2.2.4 Testing on Drones

In recent years unmanned aerial vehicles (UAVs), have become an increasingly popular in the SHM field. UAVs have the potential to revolutionise the system, due to their ability to access hard-to-reach areas, collect high-resolution data, and reduce the costs and safety risks posed by more traditional methods. Using UAVs in SHM methods reduce the need for human workers to physically access the potentially hazardous environments (civil infrastructures, offshore wind turbines, aerospace engineering etc.), and can carry various sensors allowing non-destructive inspections, producing high-resolution results. There is a rising global interest surrounding the use of UAV technology in this field of engineering, as the industry moves towards cost reduction, safety improvement, and automation in the infrastructure inspection and maintenance field.

Proof of the rapidly growing research behind the utilisation of drone technology in SHM is provided in a study by Fayyad, stating the “integration of UAVs into SHM practices has seen a significant uptick, reflecting a global trend towards more efficient and safer infrastructure monitoring solutions” [9], by leading countries such as America, China, and Europe. Further research is emerging, looking at the possible integration of newer disciplines into existing UAV technology surrounding SHM. The integration of AI would be an example of this, allowing for better detection and defect categorisation, as well as combining non-destructive testing (NDT) sensors with drone technology.

There are many technical and financial advantages to moving towards the utilisation of UAVs in SHM. The main one being the ability of drones to access high, confined and hard to reach locations that may otherwise be hazardous to inspecting personnel. Some of the less easily accessible locations that pose

a potential safety threat to workers include bridge undersides, offshore structures, tall building etc. UAVs have the potential to carry various sensors (thermal, optical, LiDAR, etc.) for the purpose of NDT, as this project aims to advance the existing research and current applications of RTI systems integrated in UAV technologies. This is especially valuable for large infrastructures such as bridges, towers, or wind turbines, where manual inspection would be time-consuming, dangerous, and impractical.

UAVs use high-resolution cameras and specialised sensors, like LiDAR and infrared thermograph, to detect and collect large volumes of data without the need for scaffolding or in-person inspectors. Khan recognises “UAVs enable efficient and high-quality image acquisition, minimising human intervention and potential errors associated with manual data collection” [35].

The integration of NDE techniques into UAV technology can be seen when Fayyad proposes a UAV structure health monitoring protocol that would inspect concrete structures using a high-resolution imagery system and automated image processing algorithms. Their results showed that drones could “effectively identify and map cracks in large structures with minimal human intervention” [9], providing a safer method for damage detection.

As well as providing many advantages within the SHM field, UAVs have many limitations which could influence the success of integrating these technologies on a wider scale. The flight time of the UAV is limited by the battery constraints such as the size and weight of the battery allowed on the drone. Unsatisfactory flight times result in the UAV not being able to complete a complete scan of the structure before power loss. This can be a significant problem when scanning large infrastructures, as the drone will need enough time to circumnavigate the structure. The limited flight time, usually up to 30 minutes for a fully equipped drone with a payload, will restrict the area the drone will be able to scan.

The constraints to UAV operations include “payload limitations, environmental sensitivity, and evolving regulatory frameworks” [9]. The payload capacity of UAVs results in many challenging restrictions for the technologies’ integration into the field of SHM. The payload capacity restricts the type and number

of sensors that can be mounted onto the drone without adding significantly to the weight or compromising stability and endurance. Most commercially used NDT sensors, like thermal cameras or LiDAR systems, would add significant weight to the drone, resulting in unstable flight control and reduced flight time.

This design reports aims to look at performing SHM tests on drones both offshore and on land. However, drones are highly sensitive to environmental conditions, such as wind, rain, and lighting. These environmental factors can cause the drone to suffer from unstable flight paths, GPS signal loss, and sensor interference. These conditions will have a negative effect on both the drones navigational and flying ability, and the success and accuracy of the sensors when capturing data from the outdoor structures.

Infrared thermography and ultrasonic testing are two widely used NDE methods in SHM. While both have proven effective under controlled conditions, they still have notable limitations when used during real-world applications. “Infrared thermography is often limited by environmental temperature changes, while ultrasonic testing is time-consuming and contact-based” [12].

Recent advancements in artificial intelligence (AI) and machine learning (ML) have catalysed a significant transformation in drone-based SHM, enabling a shift from conventional visual inspections to intelligent, automated systems. Outlined by Fayyad, “drone-based SHM has evolved into a multidisciplinary research area, where UAVs are increasingly integrated with advanced technologies such as deep learning, digital twins, and Internet of Things systems” [9]. This evolution facilitates automated data analysis, damage classification, and long-term infrastructure monitoring, reducing reliance on manual post-processing and interpretation.

Altabey and Noori highlight the growing role of AI in SHM, where machine learning methods support system identification, feature extraction, and damage detection. “AI-enabled UAVs now offer real-time defect detection and decision-making capabilities, which have proven particularly valuable in dynamic environments” [36]. They emphasise how AI can

process large datasets from sensors and imagery in real time, improving predictive accuracy under complex conditions. Deep learning applications, such as 3D shadow modelling can demonstrate the shift toward more efficient and intelligent SHM systems. These technologies allow UAVs to interpret data on-site, offering a more responsive alternative to traditional NDE methods. Highlighted by Spencer Jr., “AI-driven approaches such as convolutional neural networks (CNNs) are being used to analyse vision-based measurement systems and video feeds for identifying displacements or anomalies across large surface areas” [37].

Drones are transforming SHM by providing safer, more efficient, and cost-effective alternatives to traditional inspection methods. Their ability to reach difficult areas and capture high-resolution data enables thorough, non-contact assessments. The integration of AI and machine learning is further enhancing these systems, allowing for real-time faulty detection and predictive maintenance. Ongoing research addresses the current challenges with the technology (flight time, payload limits, environmental safety etc.). The shift towards intelligent, autonomous UAVs marks a move from manual, reactive inspections to proactive, data-driven asset management.

2.3 REFLECTIVE TRANSFORMATION IMAGING

2.3.1 Light Reflectance

Light incident upon a surface, results in both absorption and reflection of the energy transmitted. The exact behaviour of light-surface interaction is dependent upon several factors, including the material properties of the surface, as well as the angle of incidence of the light and its wavelength. Smooth surfaces give rise to specular reflection, where reflected rays remain parallel, and uneven surfaces yield diffuse reflection, with reflected rays breaking form and overlapping.

In practice, these two phenomena often occur simultaneously, with greater specular reflection creating a glossier appearance, and greater diffuse reflection a more matte appearance to the surface. To fully describe the complex reflectance behaviour created by a surface, the Bidirectional Reflectance Distribution Function (BRDF) is defined. This encompasses the direction of incident and reflected

light, given by their azimuth and elevation angles, as well as the wavelength of light.

2.3.2 Multi-Light Image Collections

Reflection Transformation Imaging (RTI) is a computational imaging technique that captures the light reflection behaviour of a surface when light is incident from multiple directions whilst a camera remains fixed in position.

This is enabled by the creation of a Multi-Light Image Collection (MLIC), which is a series of photographs taken under systematically varied illumination directions. Each photograph describes the surface response to different incident light vectors, providing data which allows analysis of reflectance behaviour.

Creating the desired illumination requires a light source to be equidistance from the subject in several positions. Highlight RTI uses a tethered handheld light source, with reflective spheres in the frame of the photograph. The role of these spheres is to calibrate the system by enabling the location of the light source to be later determined by software, using the position of the highlight on the sphere's surface to determine the angle of incidence. Alternatively, a fixed hemispherical lighting arrangement can be used, with the lights spaced at known points on the surface of the hemisphere. To reduce or remove the need for many lights, mobile lighting can be used, such as with a rotating mechanism which draws the lights sources to pre-determined locations, constructing the same hemispherical geometry. This can also be achieved with a multi-axis robot or drone. In these cases, where the light source location is accurately described, the need for reflective spheres is removed, but can serve useful for ensuring correct calibration.

2.3.3 Mathematical Methods

An MLIC created from the method previously outlined can be processed to create a per-pixel reflectance model. In essence, this is an interactive image which can be relit from any direction.

The reflectance at each pixel (u,v) under a given lighting condition is modelled by the biquadratic polynomial:

$$L(u, v) = a_0(u, v)l_u^2 + a_1(u, v)l_v^2 + a_2(u, v)l_u l_v + a_3(u, v)l_u + a_4(u, v)l_v + a_5(u, v)$$

Where:

$L(u, v)$ – Predicted luminance under light of direction

l_u, l_v – Unitless components of the normalised light direction vector

$a_0, \dots, a_5$ – Polynomial coefficients for each pixel l_u and l_v are given by:

$$l_u = \frac{x}{\sqrt{(x^2 + y^2 + z^2)}} \\ l_v = \frac{y}{\sqrt{(x^2 + y^2 + z^2)}}$$

The polynomial coefficients $a_0, \dots, a_5$ are found for each texel from solving the following system of equations using singular value decomposition (SVD):

$$A \cdot a = L \\ \begin{bmatrix} l_{u0}^2 & l_{v0}^2 & l_{u0}l_{v0} & l_{u0} & l_{v0} & 1 \\ l_{u1}^2 & l_{v1}^2 & l_{u1}l_{v1} & l_{u1} & l_{v1} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ l_{uN}^2 & l_{vN}^2 & l_{uN}l_{vN} & l_{uN} & l_{vN} & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_5 \end{bmatrix} = \begin{bmatrix} L_0 \\ L_1 \\ \vdots \\ L_N \end{bmatrix}$$

These coefficients are re-calculated for each texel, with the matrix remaining constant. From this, the luminance vector **is** found, enabling the luminance under light from any direction to be predicted at each texel.

To account for colour, an unscaled colour is computed per texel, modulated by the luminance model.

$$R(u, v) = L(u, v)R_n(u, v) \\ G(u, v) = L(u, v)G_n(u, v) \\ B(u, v) = L(u, v)B_n(u, v)$$

This is the mathematical method behind the first Polynomial Texture Maps, as developed by Malzbender at Hewlett-Packard Laboratories [38], which formed the basis of present-day RTI scanners.

Several other mathematical processing methods have been developed, with varying efficacy. The Cultural Heritage Imagine foundation provides an open-source software “Relight” for creating and viewing on the web, relightable images (RTI). This supports the traditional Polynomial Texture Mapping (PTM) and Hemispherical Harmonics

(HSH), and incorporates newer approaches developed by Ponchio [39]. A1 which are based on a joint interpolation-compression scheme, developed for web-based viewing at reduced computational cost. These approaches include Radial Basis Functions (RBF) and Bilinear Interpolation, in combination with Principal Component Analysis (PCA) data reduction.

2.3.4 Existing Uses of RTI

Reflection Transformation Imaging is conventionally applied to cultural heritage and archaeology sectors to non-invasively capture and analyse fine surface details on artifacts. This technique is particularly effective for examining inscriptions, coins, bas-reliefs and other textured surfaces through enabling interactive relighting, which reveals features often invisible under typical fixed lighting. RTI supports conservation, interpretation, and public access to cultural materials, and has become an invaluable tool for both digital preservation and research [40, 41]. In this project, we employ RTI techniques to analyse material damage and defects in wind turbine blades.

A study conducted by Coules et al used an RTI scanner to examine failure surfaces in structural materials and components, including a sheared spring ball joint and the oxidised fracture surface of a steel coil spring. This work used macro lenses on a camera positioned at the apex of a hemispherical array of 48 LEDs and employed a Hemispherical Harmonics (HSH) mathematical model to process the MLIC generated in the Cultural Heritage Imaging Foundation's Relight software, frequently making use of the built-in specular enhancement feature. The authors' setup was demonstrated to distinguish surface features in an approximate range of 10 μ m to 10mm and managed to reveal and identify details that could otherwise not be detected using conventional macrophotography, such as crack markings hidden by corrosion and shear directionality. The findings of this study, whilst not directly comparable to the project's intended application of fibreglass wind-turbine blades, demonstrate the efficacy of RTI for displaying fine details in material surface damage using open-source software, and provides some guidance on which software features are the most capable of enhancing these.

3.DRONE DESIGN

3.1 INTRODUCTION

This section details the design and development of a custom-built quadcopter tailored for the SHM inspection of wind turbine blades using RTI. The design process was driven by the need for stability, precision positioning, and lightweight integration of scanning equipment, all within the constraints of limited payload capacity and flight duration. Key subsystems, analysed below, were selected and configured to enable both autonomous and manual control modes for a more efficient scanning process. Using sensor integration and optical flow tracking, the drone can achieve the positional accuracy required for effective RTI scanning. Presented below is the rationale for component choices, control strategies, and the integration of technologies that enable high-resolution, drone-mounted surface inspection.

3.2 SCANNING PROCEDURE OVERVIEW

Based on the requirements of scanning a wind turbine blade, the following steps were outlined, shown in figure 3:

1. Drone takes off and is controlled manually through FPV until latterly aligned with the desired scanning area.
2. The Flight mode is switched to a position-holding mode, such as loiter in Ardupilot, keeping the drone stationary to ensure alignment is correct.
3. The flight mode is switched to an autonomous function that will detect the distance between the drone and turbine blade in front of it and adjust the drone's x -position accordingly to create the correct separation distance from the blade.
4. The drone returns to its position-holding mode, maintaining its distance from the turbine as scanning proceeds.
5. Once scanning is complete, the drone is switched to a manual mode and returned to the pilot.

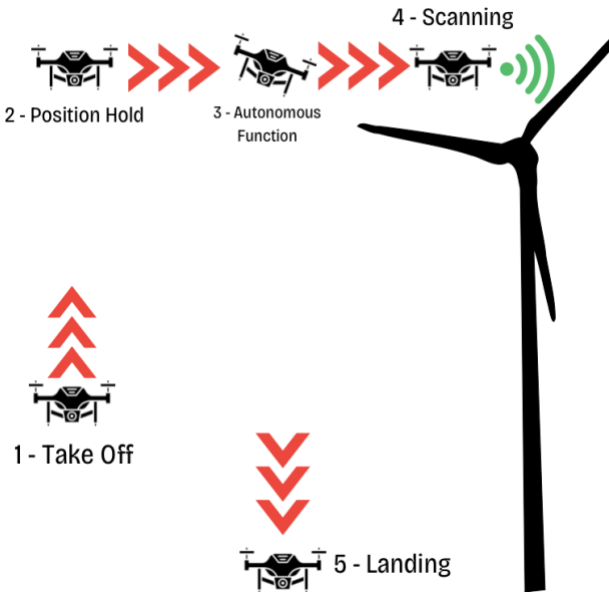


Figure 3: Diagram of the Scanning Process.

The spatial orientation of the drone is represented using a right-handed Cartesian coordinate system shown in figure 3. In this system, the longitudinal axis (x-axis) runs from the nose to the tail of the drone and is the axis about which the drone rolls. The lateral axis (y-axis) runs from the drone's left wingtip to the right wingtip and is the axis about which the drone pitches. The vertical axis (z-axis) runs vertically through the centre of the aircraft and is the axis about which the drone yaws. The three-dimensional framework used for the drone is consistent among standard aerospace conventions, being the reason this axis type was chosen as the preferred reference frame.

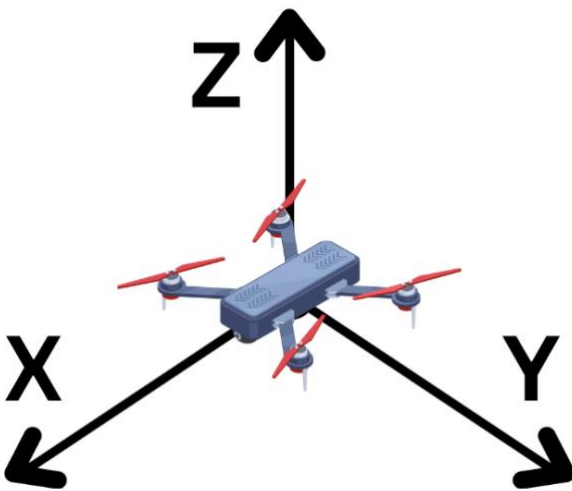


Figure 4: Spatial orientation frame for the drone.

3.3 FLIGHT CONTROLLER AND ELECTRONICS ARCHITECTURE

The foundation of the authors' design is the Matek H743 Slim flight controller, named as such after its use of an STM32H7 microprocessor, which in recent years has provided a platform for computationally intensive tasks to be done on drone. This was outlined by Julian Moosmann, Marco Giordano, Christian Vogt, Michele Magno [42], who demonstrated the ability for such microprocessors to process real-time object detection during flight. Furthermore, the H743's high memory, extensive IO and support for onboard SD cards enables the use of many external sensors and the ability to control how they affect the drone through scripting.

The flight controller communicates a desired motor output to the Electronic Speed Controller (ESC), which in turn powers the motors and acts as the interface for the battery pack to the rest of the system. A SpeedyBee 60A 4in1 ESC was chosen for this purpose, as it contains the four ESCs required for a quadcopter on a single board. Furthermore, its 60A limit is higher than both the conventional and the required amperage, allowing for the sustained power output that carrying a payload would entail.

3.4 PROPULSION AND POWER CONFIGURATION

Due to budgetary constraints, the authors elected to use a 5" propeller diameter drone, shown in figure 5, which constrained motor size, with the final design being to use 2207 size brushless motors in figure 6.. These were chosen at the lower available rating of 1800KV, which refers to the motor RPM when 1 volt is applied. By picking a motor with a lower KV value, the torque constant under high load, such as a when a payload is applied, will be reduced, thus less resistive heat will be generated which typically is a key issue under sustained high throttle scenarios. By picking a lower KV motor, a greater efficiency is achieved at an equivalent torque. Carbon fibre was chosen as the frame material due to its high rigidity; this prevents resonance in the drone leading to oscillation that can have an adverse effect on the drones control systems.



Figure 5: Aerial view of the drone with propeller diameters.

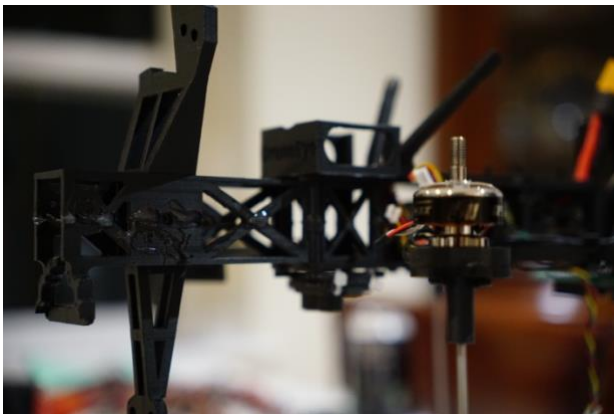


Figure 6: 2207 size brushless motors on the final design drone.

All systems relating to drone function receive power from the ESC, which itself is connected to a 1300mah 6S liPo battery pack, chosen for its high cell count wired in series. This has a nominal output range of 25.2 to 18 volts and a discharge rate high enough to power all onboard systems, even under high throttle.

The drone can be controlled manually with its 2.4GHz receiver, however achieving the precision necessary for reliable scanning will require autonomous and semi-autonomous functions. Drones can move with 6 degrees of freedom and, as such, reliable data about the position and orientation of each degree will be required. This is achieved through a combination of sensors built into and attached externally to the flight controller.

3.5 SENSOR SUITE AND POSITION ESTIMATION

To ensure stable flight and accurate scanning during inspections, a range of onboard sensors were for their ability to measure position, altitude, and movement, and then integrated into the final design of the drone. Table 2 outlines the key sensor types used in the final design, assessing their functions, mounting locations and limitations. These factors inform the selection of components best suited for a drone-mounted surface inspection system.

Sensor Type	Part Name + Mounting	Axis Sensor measures	Sensor Accuracy	Notes
Inertial Measurement Unit	• MPU6000 (accelerometer + gyroscope) • SMC on Flight Controller	• X axis + Y axis +Z axis acceleration • Roll + Pitch + Yaw velocity	• Potentially accurate however heavily prone to drift	• Only used for relative motion. Not absolute position in space
Barometer	• DPS310 • SMC on Flight Controller	• Z position above sea level (by measuring atmospheric pressure)	• Accurate within 0.5m [43], reliable but prone to noise	• Susceptible to sudden environmental noise from varying wind speeds
GPS + Compass	• TBS-M10Q: Ublox UBX-M10050 GPS + QMC5883 Magnetometer [44] • Mounted above drone	• GPS: Absolute position • Magnetometer: Absolute rotation	• GPS: low accuracy within 1.5m [44], self-correcting • Magnetometer: high precision only under	• GPS+Magnum eter allow for autonomous flight and provide frame of reference to other drift prone sensors
Optical flow + Rangefinder 1	• MTF-01: PMW3901 optical sensor + Time-of-Flight LiDAR [45] • Mounted downwards facing on drone underside	• Rangefinder: Absolute position above ground • Optical flow: linear motion relative to fixed points on ground	• Rangefinder: high accuracy (5-8m range), not prone to drift • Optical Flow: high accuracy, not prone to drift, 100Hz	• Optical flow algorithms run on sensor which outputs precise motion vectors. Tilt of drone must be accounted for
Rangefinder 2	• TF-Luna: Time-of-Flight narrow beam Lidar module [46] • Mounted forwards facing on drone front	• Absolute distance from any object directly in front of drone	• High accuracy (8m range) and not prone to drift	• Narrow beam of rangefinder used to measure an exact point. Tilt of drone must be accounted for

Table 2: Accuracy comparison table of sensors used in the final design.

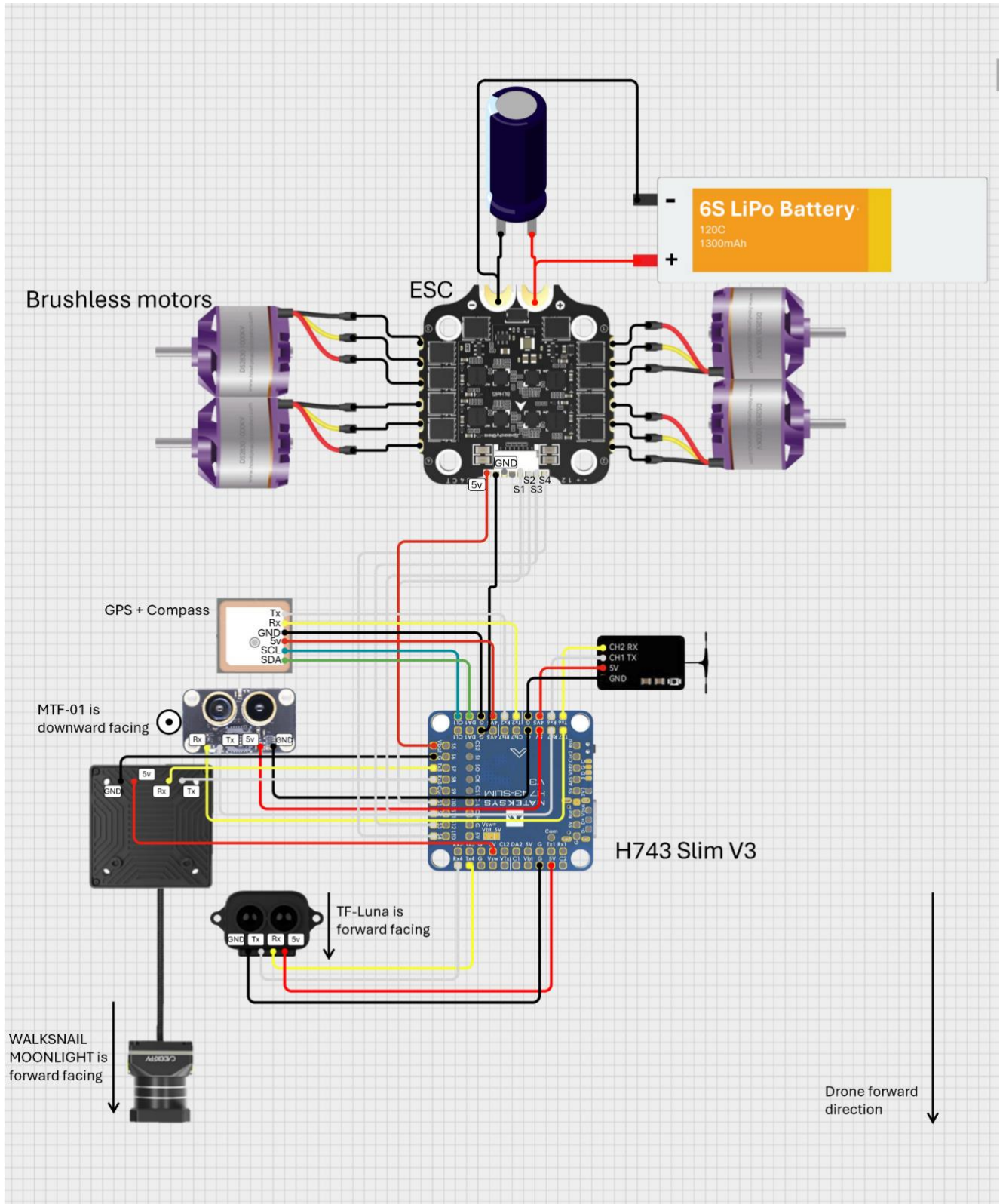


Figure 7: Circuit diagram of the drone control.

3.6 OPTICAL FLOW AND INSIDE-OUT TRACKING SYSTEM

This drone will use GPS tracking to give a general estimate of its position, like prior larger scale quadcopter scanning techniques. A novel inside out tracking solution will be used to build on this position estimate to more accurately orient the drone using the MTF-01 optical flow / lidar system

alongside the TF-Luna lidar. To provide positioning information to a sub-meter accuracy that a GPS could not obtain in isolation, the MTF-01 was chosen only as an inexpensive lightweight optical flow sensor example that has a rangefinder built in. Higher range sensors are also available or able to be constructed as outlined by prior authors [47].

Optical flow was chosen due to its relative popularity as an inside out tracking solution for UAVs as well as being a well-established baseline for a camera-based positioning solution. The system works by taking consecutive images of a surface and assuming the time difference between them is small. A Lucas-Kanade optical flow algorithm is then used to generate velocity vectors as follows:

The sensor generates sequential monochromatic images of the ground. The intensity of the luminance of each pixel in the image is recorded, compared with the prior value and its change in displacement over time (δt) is found and recorded as

$$I(x, y, t) = I(x + dx, y + dy, t + dt)$$

which can be expressed using a Taylor expansion as

$$I_x u + I_y v + I_t = 0$$

This is known as the optical flow constraint equation and contains unknowns of only u and v per pixel. To obtain a solution, a 5x5 window around the pixel is considered, in which all other pixels are assumed to have the same motion.

For n pixels in the window, the constraint equation is as follows:

$$\begin{bmatrix} I_{x1} & I_{y1} \\ I_{x2} & I_{y2} \\ \vdots & \vdots \\ I_{xn} & I_{yn} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} I_{t1} \\ I_{t2} \\ \vdots \\ I_{tn} \end{bmatrix}$$

This can be solved for u and v using the normal equation

$$A^T A \cdot \begin{bmatrix} u \\ v \end{bmatrix} = -A^T b$$

where $A^T A$ is invertible.

It is evident the limitation of this algorithm is the assumption that adjacent pixels retain their luminance across frames and the potential for inaccurate values of u and v to be derived at high velocities. However, these compromises can be accepted for a prototype concept to assess the validity of outside-in optical solutions to improve tracking. This has been used by prior authors Noel P. Caliston, Chris Jordan G. Aliac and James Arnold E. Nogra [48], to achieve sub-meter accuracy on the order of 10-20cm, albeit with variance based on

flying conditions. However this far exceeds the 1.5-3m accuracy of conventional drone GPS modules [44].

3.7 SENSOR FUSION AND STATE ESTIMATION

These velocity values are calculated on the sensor and sent as vectors to the flight controller, alongside positional data from all other sensors. These are combined by the Ardupilot firmwear running on the Matek H743 using an Extended Kalman Filter 3 (EK3) into a unified state estimate. This represents the drone's best estimation of its position, velocity and orientation in 3D space, calculated using a dynamic motion model that probabilistically weighs each sensor input according to its accuracy and noise. The EK3 fuses inputs from the IMU, GPS and compass unit, barometer, rangefinders and optical flow sensors, filtering out noise and correcting for biases in real time. This estimate is continuously updated and used in the flight control algorithms to stabilise the quadcopter and execute other autonomous functions with precision. Crucially, the STM32H7 processor of the Matek H743 is able to compute this without perceptible latency.

Despite its lack of accuracy, the GPS is still used as the foundation of the EK3 positional estimate, from which the MTF-01 will more precisely orient itself relative to the ground. The TF luna will subsequently orient the drone at a correct distance to a scanning surface using a custom LUA script.

3.8 CAMERA SYSTEM AND SCAN STABILITY

RTI scans do not allow for significant motion between images, and the field in which subsequent images can be overlapped is small. As such, outside of improving positional accuracy of the drone, the only way to improve the quality of scans is to use a stabilised high-resolution camera seen in figure 8. As the scanner is mounted in front of the drone, the camera must be as lightweight as possible as to not further imbalance the drone's mass distribution. This has been labelled by prior authors [49], as a crucial aspect of improving drone control. However, it is mentioned that the battery can be used as a counterweight, which is incorporated in the design.



Figure 8: High resolution camera used in the final design.

Recent advancements in the miniaturisation of high megapixel camera sensors and image processing chips have given rise to a series of isolated camera sensor and recorder assemblies that are designed to be mounted on lightweight platforms such as drones. The WALKSNAIL MOONLIGHT is one such example, having a sensor capable of recording in 4K at a high bitrate of 150mbps while weighing only 38.5 grams – less than half the weight of an equivalent GoPro. The camera receives power via the flight controller and records on an onboard SD card while transmitting a real-time video feed to a pair of FPV goggles. These allow for remote manual navigation and provide basic telemetry such as battery life and altitude.

3.9 FINAL DRONE SPECIFICATION AND JUSTIFICATION

The flight performance of a fully assembled drone is heavily dependent on its mass to thrust ratio with a maximum acceptable value for larger drones being 1:5 [50]. This ratio as well as the distribution of mass across the drone, and its maximum thrust also affects the final PID tunes required. With calculators for

these being readily available for Ardupilot. [51]. This resulted in the author's unloaded prototype drone having the following specifications:

Total Mass (Unloaded)	617g
Dimension (Unloaded)	226mm
Battery Capacity	1300mAh
Maximum Thrust	7600g
Mass: Thrust Ratio (Unloaded)	1:12
Hovering Current Draw (Unloaded)	13A
Maximum Current Draw	60A
Estimated Loaded Flight Time (Unloaded)	8 Minutes
Inside-Out Tracking Range	8m

Table 3: Specifications of the unloaded drone.

These values represent the limitations of a prototype under budgetary constraints for the primary purpose of assessing the viability of outside in tracking solutions in obtaining RTI scans

4. RTI SCANNER DESIGN

4.1 INTRODUCTION

Given the previously mentioned criteria, a prototype RTI scanner must be combined with a 5-inch propellor drone to conduct scans on the surface of wind turbine blades. As a prototype the design will be created to demonstrate how an RTI scanner should be added to a drone given considerations and limitations mentioned in the literature review and drone design.

4.2 DESIGN CONSIDERATIONS AND REQUIREMENTS

There are three key design considerations that the RTI design must adhere to and optimise: Functionality, mass and drone compatibility.

Given the use of a drone with 5-inch propellers, the maximum thrust generated is 7.6 kg. Given the assumption that the thrust to mass ratio should be at least 5:1 this limits the mass of the drone to 1.52kg [52]. For drone control it is essential that the distribution of mass is as even as possible with the centre of mass being as close to the centre of the propellers as possible.

Lighting must be provided along the surface of a hemisphere with maximum spaced radial angles of 45 degrees and azimuthal angles of 15 degrees (30 around the camera). The changes in light picked up by the camera must be from the RTI and not external interference. The camera must be located at the top, as to get an effective scan. An analysis of RTI scanning highlighted the need to “move the light source to an approximate position in a flexible sampling pattern before each shot, to create a hemispherical set of lighting samples around your photographic subject” [53], for an effective scan.

The design of the RTI must accommodate attachment to the drone without interfering with drone functions. As the RTI is attached to a drone it will experience high accelerations in all directions leading to accompanying forces on the RTI. Structural and dynamic stability is therefore essential. A study on FPV drone structural analysis emphasises the “use of lightweight and high-strength materials such as carbon fibre can significantly reduce the weight of the drone and increase its structural integrity, resulting in improved stability and manoeuvrability” [54].

There are also performance design considerations that should be accounted for to optimise performance. Speed of scans to allow for a scan comfortably within a flight time. Scan area must be large enough for significant coverage from sampling, this is determined by the radius of the lights and camera. Adaptability means that changes on sight can be made to account for changing conditions that may be expected like ambient light or visibility issues. Weather resistance means that it will be able to resist high winds and sudden rain for short periods.

4.3 CONCEPT DEVELOPMENT

The RTI attached to the drone must be changed from a conventional dome design as it would act as a sail and cause interference to the drone’s performance. A possible alternative considered would be to use multiple drones where one acts as a camera drone while the others act as light sources. Alternatives such as multi-drone RTI systems, including the previous design, were dismissed due to the impracticality of the lighting positions needed. This would result in slow operation or complex drone coordination that would be overly convoluted.

The final RTI design replaces the dome with a moving arm. This introduces more susceptibility to inaccuracies from external lighting however does allow for more lighting angles and adaptability. This approach is consistent with RTI methodologies where “image capture for RTIs is performed using a fixed camera and either a lighting array or a single light source which can be moved around the subject in-between shots” [55]. An arm design would substantially reduce the mass and have much better compatibility with a drone than a dome RTI design. A study which introduces a “robotic arm-based RTI system designed for capturing images of cultural heritage surfaces: LightBot” [56], has shown that a moving arm RTI is possible and provides accurate results.



Figure 9: Traditional RTI dome method 1.



Figure 10: Traditional RTI dome method 2.

The primary consideration attaching the RTI arms to the drone is the position and method they would attach. The most basic ways to attach would be either below or in front of the drone. This is shown in figures 9 and 10 using a traditional RTI dome to show the arm paths. Attachment below ensures mass is centrally distributed and would be stable, however concerns in take-off and landing and the limitations

on the RTI radius make it impractical. Attachment in front allows much more control over the design of the RTI and means the scanner can be near the scan surface. Attachments in front mean the distribution of mass is uneven and would require the introduction of counterweight measures, increasing overall weight. This design also requires the movement of the drone camera and position sensor to the front of the attachment to avoid interference.



Figure 11: Forward facing arms RTI design 1.

To overcome the issues of the in front design a central attachment was proposed, meaning mass distribution is more even and the camera and position sensor can remain in place. As seen in figure 11 the use of this setup requires the arms to accommodate the drone's blades by curving around them before becoming semicircular beyond the blades. This design would require arms with an over meter diameter to overcome the significance of the exclusion zone in front of the drone. The drone blades would be spaced to accommodate the arms passing between, reducing the thrust per area of the drone. Overall, these drawbacks mean the design would not be possible in practice and an in front attachment must be used.

There are two possible mechanisms for the arm rotation: direct position control using a servo motor attached to a shaft to rotate the arm or using a circular track and wheels for the arm to rotate around. Given the need to balance the mass, the design must use two arms with opposing motion to maintain the centre of mass and ensure no net torques are acting on the body of the drone. The design must also leave a central space to accommodate the drone camera and secondary rangefinder.



Figure 12: Forward facing arms RTI design 2.

The use of a basic motor attached to a shaft mechanism would be the most practical given the rotation of a single arm and without having to house the drone components. However, the introduction of a second arm means that the design becomes as complex as using a track mechanism. This mechanism would also require a servo motor or stepper motor capable of high torques which would substantially add to the overall mass.



Figure 13: Circular rail track system.

The use of a track system allows for the easy accommodation of components in the centre of rotation and the use of lower torque motors. This, therefore, was the mechanism used. The track system can either work with a chain/belt pulley or motorised wheels. Given the components must be as small as possible for mass considerations, the use of motorised wheels is not feasible. A belt pulley is used to save on mass compared to a chain and due to the belt's scalability.

4.4 FINAL RTI DESIGN



Figure 14: Rotating arm RTI final design.

The final design for the rotating arm RTI is shown in figure 14. This displays the track pulley mechanism to rotate the arms and how it attaches to the front of the drone. The camera attachment component is shown at the centre of the arms with the position hold sensor being located above it. A minimum RTI diameter of 500 mm was chosen to meet the lighting distribution constraints while minimising aerodynamic drag and mass, given the size of the drone selected and ease of initial testing,

To allow for iterative design changes all custom components are manufactured using 3D printing and parts are connected using of M4 nuts and bolts. Lightweight flanged and small wheel bearings and a 50g continuous servo motor was selected for minimal mass and reduced friction (see Appendix X for parts list). Given many of the 3D printed components will act as load bearing parts they were constructed with carbon reinforced PLA to improve strength and structural rigidity, compared to conventional PLA. To reduce the weight during flights, several bolted joints were replaced with epoxy bonding.



Figure 15: Track Mechanism.

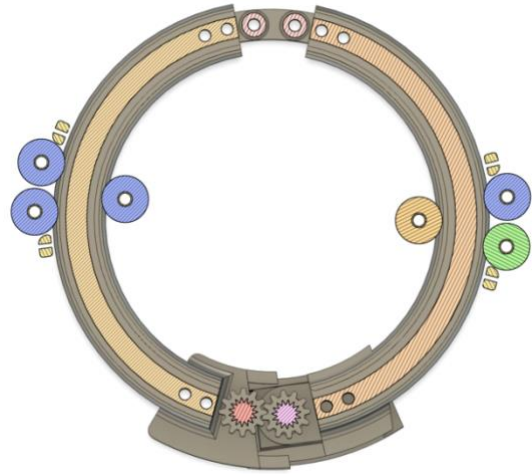


Figure 16: Cross section of track mechanism.

The track mechanism and its cross-section are displayed in figures 15 and 16, showing all components, the positioning of the wheels and the channels where the timing belts will pass through. Two separate tracks are used rather than a single circular one for individual arm control. Belt tension and gear alignment were improved to ensure smooth arm rotation, in addition to synchronised rotation, highlighted in figure 16. The ends of the timing belts are secured into the arms, moving the wheels along with the belt. One of the belt gears is directly attached to the servo motor while the second belt gear is fixed in place with bearings to ensure the motion of both arms will oppose each other exactly.

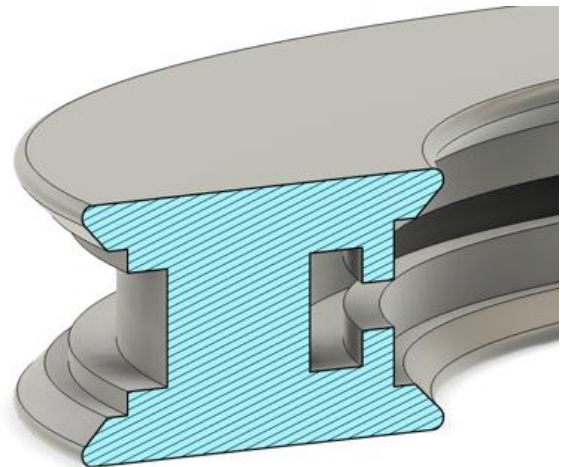


Figure 17: Cross sectional area of the track.

To allow placement as close to the drone body as possible and reduce the mass, each component was made as small as feasible. The first component of consideration is the track whose size was limited by the size of the wheels and timing belts. The track

dimensions were minimised based on component constraints, resulting in an 11 mm cross-section height with internal channels for belt routing and a centre core width of 7mm, to ensure structural stability demonstrated in figure 17. The inside and outside trenches accommodate the timing belt and prevents tension pulling the inner belt section straight. Given the size of the components required at the end of each track, each track covers 155 degrees.

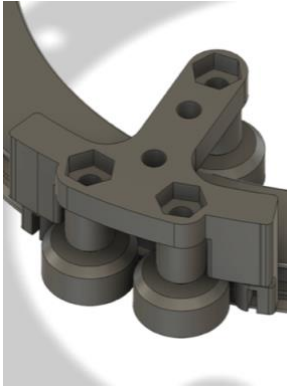


Figure 18: Mass efficient design 1 of the wheel and track mechanism.

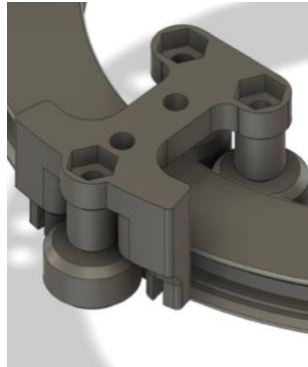


Figure 19: Mass efficient design 2 of the wheel and track mechanism.



Figure 20: Final design - belt and component connection.

A component, consisting of three wheels and two-timing belt connection points, is created to join the track wheels to the belt and RTI arms. The timing belt connection points must be located on the outer edge of the track as the belt is enclosed on the inner edge of the track. The final arm mount design, in figure 18, was chosen for its lower mass and smaller angular range, allowing for more compact LED placement. The connection between the belt and the component, in figure 19, ensures a compact design.

The connection points between for the belts are curved to avoid collision with the wheels where all parts are given a clearance of 1mm.

There was substantial friction from 3D printed surfaces and many points of contact between moving rough surfaces, therefore design adjustments were made to reduce drag and improve efficiency. The mass of the servo and subsequently the torque required were kept as small as possible. Resulting torque analysis determined a torque of 0.54 Nm would be sufficient to deal with the inertial forces and torques experienced (Appendix X). ms^{-1}

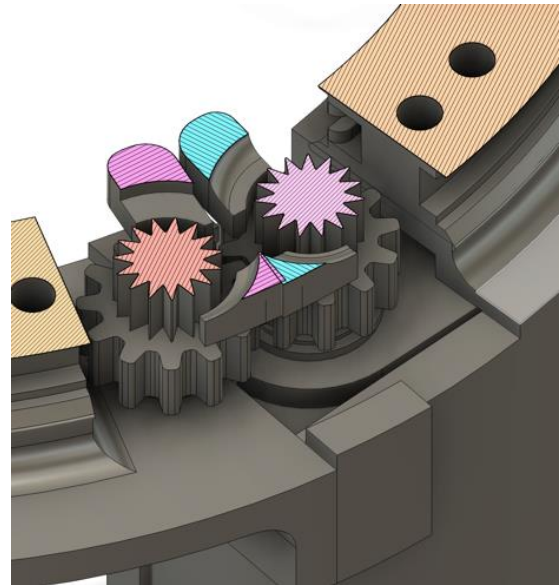


Figure 21: CAD view of the arms holding the belt against the belt gear.



Figure 22: Photo of the arms holding the belt against the belt gear

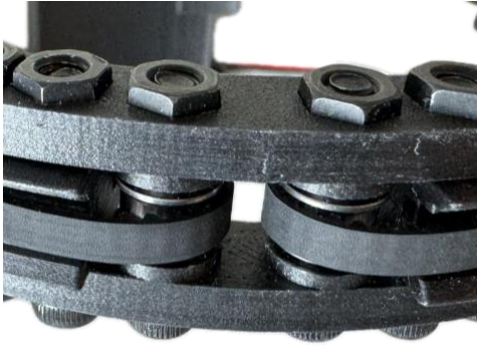


Figure 23: The belt passing over the ends of the tracks

Slack in the timing belt, when stationary, was intentionally added to reduce belt friction during movements and reduce gear bindings. Arms were added over the belt gear teeth holding the belt against the belt gear, to prevent belt skipping and allowing slack in the belt, shown in figures 21 and 22. These design improvements also reduce the likelihood of damage to the belt gear from shear forces.

These components lay over the belt gear, shown in figure 22 and fix to the top of the belt gears with bearings to hold the belt gears upright. These again reduce the friction experienced by the motors shaft and ensure the lower gears of the belt gear interlock correctly to prevent unnecessary friction. Figure 23 shows how the belts pass over flanged bearings for minimal friction.

4.5 CIRCUIT AND LED CONTROL DESIGN

Each arm should contain 8 LEDs, maximising coverage, given the RTI diameter of 500 mm. The LEDs must provide a luminosity capable of overcoming ambient light due to the absence of a dome. The batteries account for a large proportion of the mass, therefore the LEDs selected must also have a low enough power draw so that lightweight batteries can provide a sufficient flight time. Resultingly, 3W white LEDs were selected

An Arduino is used to control the LEDs, connected to an external power supply, individually. An Arduino uno must be used as 2 adjointed shift registers and 3 control pins are required to control the 16 LEDs. The Arduino can control the servo motor. Each LED is mounted on a control board that uses a MOSFET transistor, allowing for the output signal from the shift registers to be connected to the gate of each MOSFET and act as a switch for an external power supply for the light.

The 16 LEDs are attached to a 3.7v lithium-ion battery and can output a current of up to 10.5A. The LEDs are wired in parallel so that even if all the LEDs are on at a given time, they would be able get sufficient current to operate at near maximum power. The servo is connected to a second battery attached to a DC/DC step up boost converter, providing the necessary 5v voltage for operation.

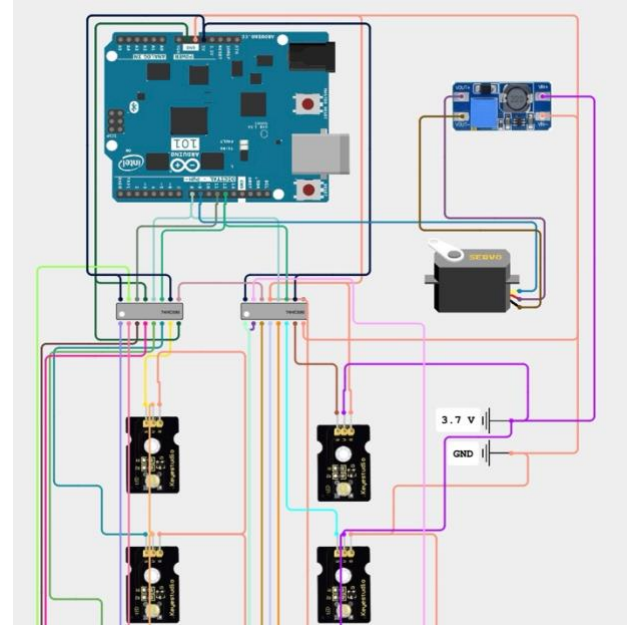


Figure 24: Servo motor and LED control circuit diagram. Wiring for all 16 LEDs and 4 LED boards shown.

The Arduino is powered by a conventional 7v battery attached to an infrared switch to allow for remote switching of the light/motor cycle.

A single Arduino allows for synchronised motion of the LED, light control, and servo motor. The continuous servo motor provides no feedback signal making the system open loop. The servos are therefore not controlled by the position of the arm but the speed of motion and the time moving. This requires manual calibration to ensure the arms are in their correct positions before each test. Calibration is easy for the user to conduct, as the top and bottom points are where the arms meet, and time steps must therefore be selected that complete this full range of motion in 6 steps (30-degree separation) or 4 steps (45-degree separation). The return to the original position can be completed in a single step. This method of control is simple and very easy to make quick changes by updating the Arduino code. Issues, however, arise when the arms are subject to unaccounted forces as the arms may not move to the exact intended position.

4.6 SELECTED RTI FUNCTIONS

Within the open-source Relight software from the Cultural Heritage Imaging foundation [57], several models are available for selection, used to process the multi-light image collection (MLIC) produced by the arm-based drone-mounted scanner. These models enable relighting by interpolation between images with varying light conditions. RBF27 was selected for our application.

For each pixel in the image, its reflectance behaviour is represented as a high-dimension vector composed of its RGB colour values across all captured images. To compress this data, Principle Component Analysis (PCA) is performed across all pixel vectors. This results in a mean reflectance vector and a set of 27 basis vectors (PCA basis vectors), from which each pixel can be approximated using a relatively small number of scalar coefficients (PCA coefficients).

When a lighting direction is selected, the system uses Radial Basis Function (RBF) interpolation to estimate the reflectance response of each PCA basis vector (and the mean vector) at that direction. The RBF weights are calculated using a Gaussian kernel based on the angular distance between the desired light and the original capture directions. The final relighted pixel value is computed as a linear combination of the interpolated mean and basis vectors, scaled by the pixel's PCA coefficients.

Ponchio demonstrated that their developed RBF27 outperforms PTM, HSH, and their Bilinear method in terms of image quality [39], when the light sources are densely and regularly arranged, which suits the arm-based lighting arrangement developed where 96 total light positions are used to capture images. Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) were the selected quality metrics used to compare approaches in their study, comparing original photographs captured with produced renderings under the same lighting conditions.

4.7 RTI SCANNER AND DRONE ASSEMBLY



Figure 25: Structure connecting the RTI to the drone body.

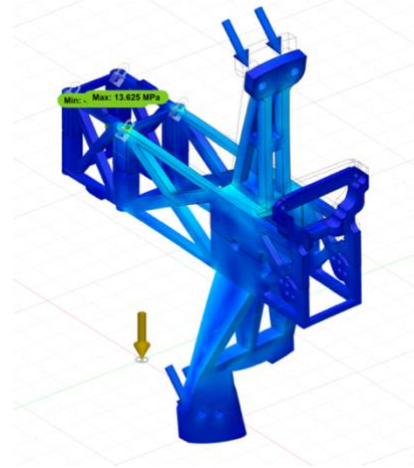


Figure 26: Stress distribution map of the structure.

Figure 25 shows the RTI assembly mounts a custom 3D-printed structure secured to the drone through the 4 bolts. This structure is then secured into the drone body by replacing the camera mount fixtures. The front of the structure contains the replaced camera mount and a housing for the position hold sensor. For CAD testing a safety factor of 4.5 was chosen, due to inherent structural flaws present in 3D-printed parts and the large accelerations that would be experienced. During load testing demonstrated in figure 26 and generative iteration, a load of 10N was applied horizontally and vertically to replicate the forces experienced during operation.



Figure 27: 3D model of final RTI arm design

All the components in the 3D model were provided with the material. It was also assumed that all components were printed with 100% infill, so the density of printed components is same as the filament used. These considerations provide a maximum estimation of mass and the location of the centre of mass of all the components collectively. The maximum mass found was 490g at 165mm from the centre of the drone body corresponding to a maximum moment of 0.79 Nm. The mass was found to align with the maximum prediction once LEDs and wiring were included at 486g.

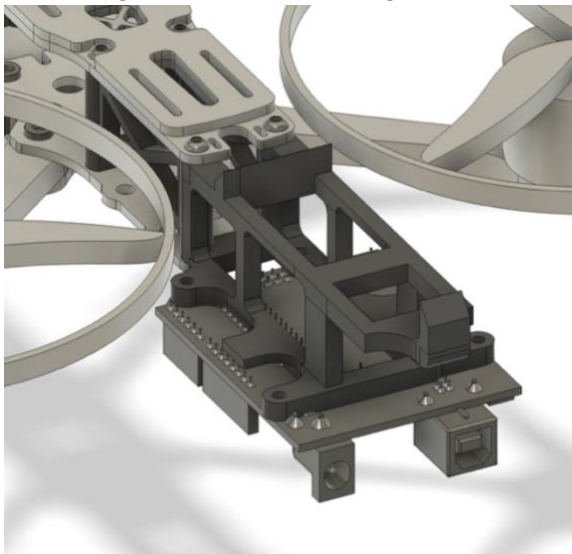


Figure 28: Rear battery and circuit board mount.

To counter the moments experienced without contributing substantially to the overall mass, a structure was created to house batteries and circuitry behind the main frame of the drone. This structure only contributes around 25g to the overall mass but provides a counter moment that moves the centre of mass of the structure to around 36mm from the lift centre, comfortably in the acceptable range.

The mount shown serves a second purpose to accommodate the Arduino. This allows for the wires to run under the body of the drone and avoid risk of being cut by the propellers. It also houses the IR switching board and the boost converter board to ensure all wiring is compact, further saving on mass and reducing the risk of wiring being cut by propellers.

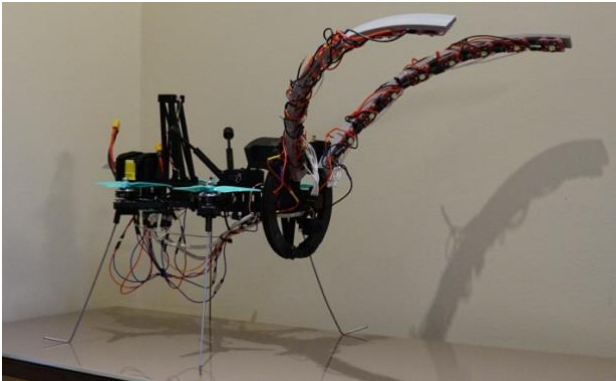


Figure 29: Final RTI and drone assembly picture.

Figure 29 displays the final assembly of all RTI and drone components. As shown the RTI control boards are mounted behind the drone's main body and wiring attached to the RTI itself passes under the drone to avoid the propellers. To ensure take-off/landing clearance for both the RTI track and the Arduino board, legs have been added. The legs consist of a small 3D printed PLA casing connecting a thin ridged steel rod with the drone body under each propeller motor.

The key loaded drone specifications are as follows:

Total Mass (Loaded)	1165g
Dimension (Loaded)	595x500mm
Battery Capacity	1300mAh
Maximum Thrust	7600g
Mass:Thrust Ratio (Loaded)	1:6
Hovering Current Draw (Loaded)	33A
Maximum Current Draw	60A
Estimated Loaded Flight Time (Loaded)	4 Minutes
Inside-Out Tracking Range	8m

Table 4: Specifications of the loaded drone

5. TESTING AND RESULTS

5.1 TESTING METHODOLOGY

5.1.1 Tests Conducted

To determine the feasibility of using the RTI drone system for wind turbine blade scanning, first a few operation criteria must be established for both the drone and the RTI scanner itself. These include speed of scans conducted, position of the drone during a scan, stability needed for each scan and acceptable testing conditions.

Initially, the parameters mentioned are determined through static RTI testing, allowing for testing to be conducted in a controlled environment. After conducting these tests, the determined conditions and parameters are used to do final static testing. These results are used to determine the quality of the RTI scans given the LEDs, arm mechanism, camera and other components used in the prototype.

Drone testing was then conducted where criteria such as flight time, stability and vibrations were evaluated, with the drone both loaded with the RTI and unloaded. This provided further insight into the impact of weight and weight distribution, and if the drone setup can provide sufficient stability to successfully conduct RTI scans. Further testing of the front rangefinder can then be performed using the established positioning from static RTI testing.

Final testing is conducted where the RTI mechanism is activated while in flight and the impact of its actuation on stability can be evaluated.

5.1.2 RTI Testing Setup

All scans are conducted while all the RTI parts are attached to the drone. For static tests, the drone body and legs act as a stable base to ensure the RTI scanner does not move during operation. Given the purpose of the scanner will be to evaluate vertical structures, all testing is conducted horizontally rather than vertically, as is the convention.

The camera used for the drone control serves a second purpose to record footage for the RTI scans. At the initiation of each scan, the camera is set to record and will capture the surface of interest through the full cycle of the scan. Each RTI arm during the scan cycle will assume 6 equidistant positions, with each light flashing for 250ms. Given the time of actuation between arm positions is known,

a frame is captured every 250ms 16 times at each position. This provides 96 images at 96 separate lighting positions, providing a large dataset to create results.

Each static RTI scan was taken using reflective spheres (marbles) to calibrate the positions of the LEDs. This ensures each test provided an MLIC with the exact accompanying light source coordinates, which also produced a light source map for evaluation of the consistency of light positions relative to the camera. The exact coordinates calculated for the light array are also given in a separate text file. All images collected are added into the aforementioned Relight software where different aspects of the RTI scan produced can be isolated and enhanced. This includes showing regions of specular reflections or diffuse reflections, which can be dynamically re-lit from any desired position. Specular results were widely used for comparisons as they display small surface details most effectively during visual evaluation. Across multiple initial tests, the optimal position for displaying variations in surface depth was ascertained and subsequent results were re-lit using the same lighting vector.

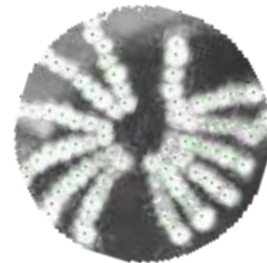


Figure 30: Example of light map produced from scan.

Scans produced by RTI scanners are inherently two-dimensional, in contrast to other imaging techniques such as LiDAR scans, and are produced to outline variations in surface depths which constitute defects. As such, it is not possible to quantify the extent of surface height variations with this technique alone, and so quantitative comparison between results is not possible. As such, visual comparison is employed to evaluate efficacy. Results are deemed acceptable if variations in surface depth exhibit substantial contrast in luminance across all visible points within the scan, without further increase from adjusting the selected test parameter.

5.1.3 Drone Testing Setup

To assess the performance of the drone in terms of positional accuracy and autonomous function, data can be collected from test flights using the built-in logging feature of ardupilot. This records the input data from all sensors as well as key EK3 system parameters and thus demonstrates the drone's ability to control itself.

To ensure controlled testing conditions for each drone flight, tests are conducted with the following criteria:

- Same weather conditions
- Reliably secure GPS lock
- Same drone components
- Consistent flight sequence
- Calibration of the accelerometer and magnetometer between tests to ensure reliability

Two types of test flight are undertaken in both a loaded and unloaded case. During the first type the drone is flown vertically in position hold and kept stationary by its autonomous systems, until fully depleting the battery and initiating the autonomous landing failsafe. During this test, data regarding the absolute position, velocity and vibrations within the drone are obtained. Vibrations can be determined from the high frequency variations in acceleration recorded by the accelerometer. This data is logged from the optical flow sensor, downward facing rangefinder, GPS and barometer + IMU on the flight controller. This will allow for an assessment of the individual performance of each sensor, as well as an assessment of the EK3's ability to combine their data to increase overall precision in stationary flight.

High levels of vibration or sway can be attributed to specific sensors or diagnosed from other features of the drone depending on the precision of motion. Notably in the loaded case, the RTI arm sequence will be activated to assess its effect on drone flight.

The second form of test flight will assess the performance of the autopilot script designed to keep the drone in scanning range of a surface, this will be primarily assessed based on the front facing rangefinder distance, as well as the drone's absolute position taken from the EK3. Oscillations in the drone's motion relative to the scanning surface are

shown by the variation in horizontal velocity over a given time. These results would help highlight latency in the drone's ability to run scripting functions, assuming other sensors are deemed accurate by the initial assessment

When evaluating the front rangefinder performance, the distance held from a surface will be 100mm further than the distance determined from static RTI testing to allow for substantial clearance given displacement variation in the x-axis. The same flight process is employed as in the first stage of drone testing, however once a stable altitude is reached, the autopilot will be activated to atomically position the drone at the intended separation distance from the surface to be scanned.

5.2 RTI TESTING RESULTS

5.2.1 Control Scanning Surface

During initial functionality checks, it was clearly demonstrated that the scans must be completely stationary beyond minor vibrations.



Figure 31: Static RTI control surface.

A matte white wall with moderate surface roughness was used as a control wall for comparing scan parameters. This wall was selected as the wall texture cannot be seen prior to scanning, however acceptable results will display substantial surface roughness.

5.2.2 Light Flash Length Test

To optimise the speed of tests, the length of time that the lights flash for should be minimised. Tests were taken at a range of flash lengths with the two displayed showing 500ms (standard length with acceptable results) and 250ms (The minimum possible length of flash).

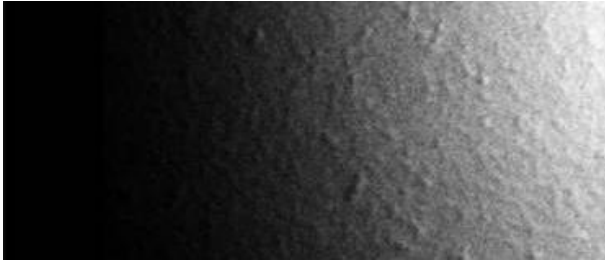


Figure 32: *Specular RTI with 500ms light blinks.*

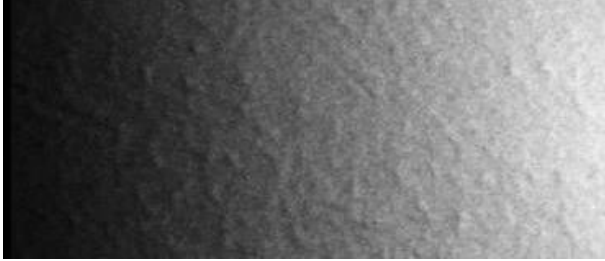


Figure 33: *Specular RTI with 250ms light blinks.*

As can be seen there is drop off in quality from taking tests at 250ms compared to 500ms. This change, however, is minimal, and both results display an acceptable resolution. 250ms appears to be the limit as recordings are taken at 24 frames per second, leaving 6 frames per light. This the smallest found number of frames for the camera to adjust to new lighting following repositioning.

5.2.3 Scanning Distance Tests

For drone testing, the distance that the camera and the front rangefinder is from a surface must be established for optimal scanning. Tests are therefore conducted at 250mm (radius of the arms), 300mm, 350mm and 400mm.

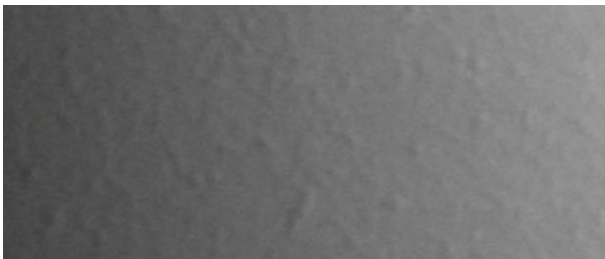


Figure 34: *Specular RTI with a camera distance of 250mm to the scanned surface (radius of the RTI arms).*

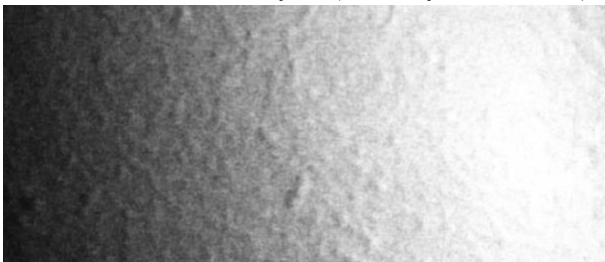


Figure 35: *Specular RTI with a camera distance of 300mm to the scanned surface.*

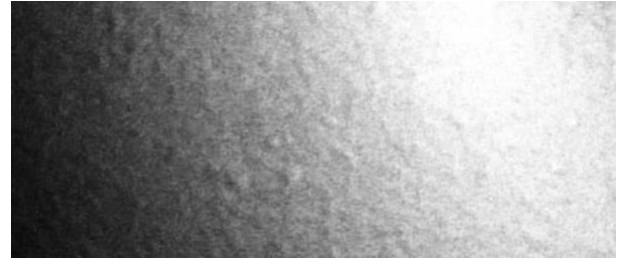


Figure 36: *Specular RTI with a camera distance of 350mm to the scanned surface.*



Figure 37: *Specular RTI with a camera distance of 400mm to the scanned surface.*

Scans at a distance less than 300mm from the camera provide usable results but not optimal. Results at distances of 400mm are unusable for textured results. Therefore, the optimal range will be between 300 and 350mm. To ensure drone clearance to avoid collision, the front rangefinder should be set to a distance 350mm for final drone results.

5.2.4 Acceptable Ambient Light Tests

The amount of acceptable ambient light must now be found to provide adequate results. Each result is therefore shown with an image of the ambient light and then the test results.

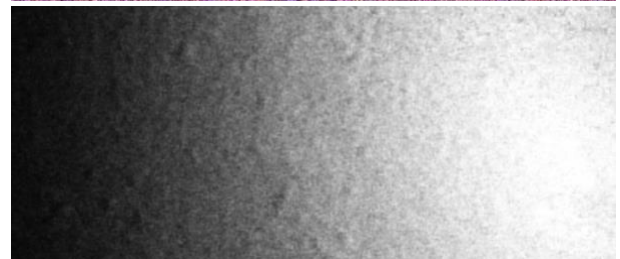


Figure 38: *Relative low ambient light RTI test prior to scan (top) and after specular scan processing (bottom)(the top image has added contrast).*

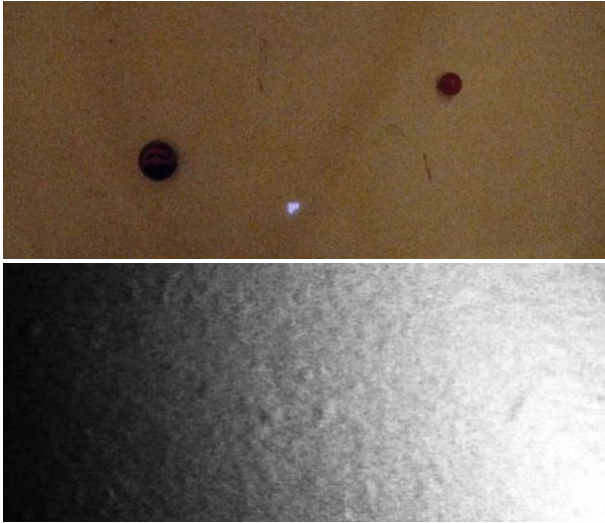


Figure 39: Relative moderate ambient light RTI test prior to scan (top) and after specular scan processing (bottom).

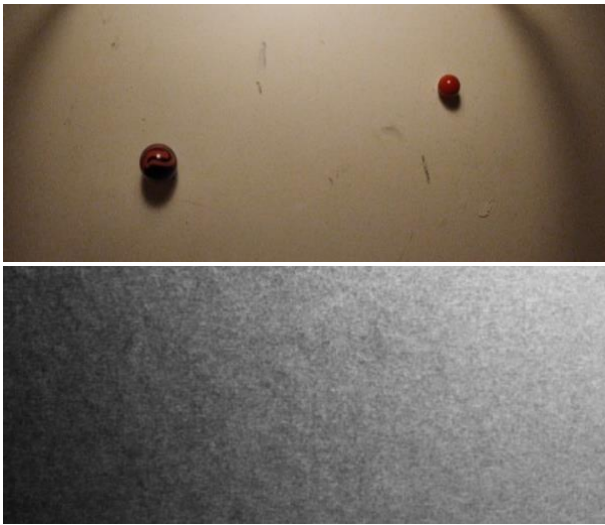


Figure 40: Relatively high ambient light RTI test prior to scan (top) and after specular scan processing.

The introduction of any ambient light will introduce noise, however moderate levels of ambient light will still provide results with acceptable levels of resolution. The RTI will therefore be able to perform to an acceptable standard after sunset even with moderate levels of light from sources like street and house lights.

5.2.5 Final Static Testing Results

Final testing was conducted on a variety of surfaces. These include surfaces with high roughness, surfaces with minimal roughness, like in previous testing, and smooth surfaces. For the smooth surface, a sheet of polypropylene (pp) with slight curvature and light scratches was used, whereas the other two tests were conducted on flat painted walls, with different roughness levels.



Figure 41: Smooth surface RTI results prior to scan (top) and after re-lit scan processing (bottom).

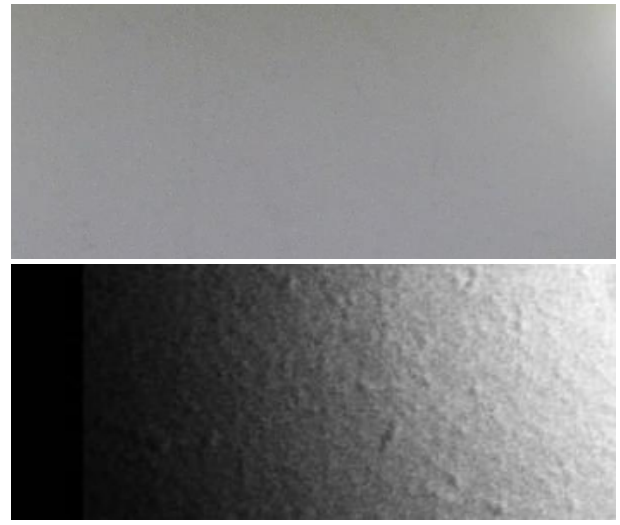


Figure 42: Moderate surface roughness RTI results prior to scan (top) and after specular scan processing (bottom).

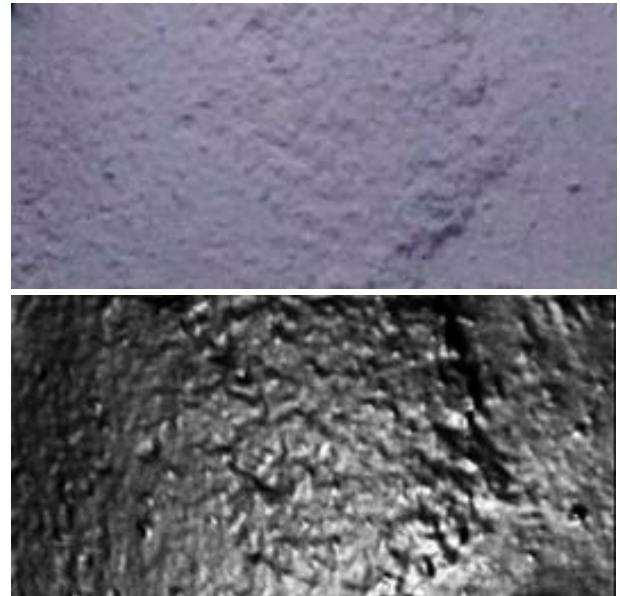


Figure 43: High surface roughness RTI results prior to scan (top) and after specular scan processing (bottom).

5.3 DRONE TESTING RESULTS

5.3.1 General Performance

The loaded and unloaded cases for the drone under stationary flight yielded the following flight times for the mass ratios of 1:12 and 1:6 respectively.

Loaded Flight Time (minutes)	7:34
Unloaded Flight Time (minutes)	1:42

Table 5: Flight times before and after loading.

Notably, under loaded flight the low voltage automatic landing failsafe would trigger while the battery had up too 40% of capacity remaining. The operating currents during hover were 33A for the loaded case and 13A for the unloaded case.

5.3.2 Stationary External Sensor Positional Tracking

The altitude of the drone was measured from the internal flight controller barometer and external downward facing rangefinder while hovering as follows:

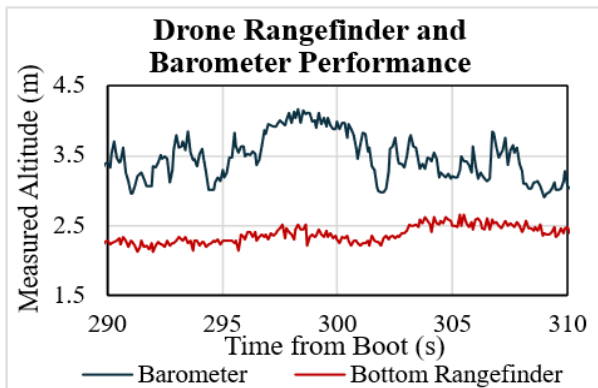


Figure 44: Graph displaying drone rangefinder and barometer performance.

During which the position of the drone was recorded from the GPS and the internal EK3's refined estimate using the optical flow sensor:

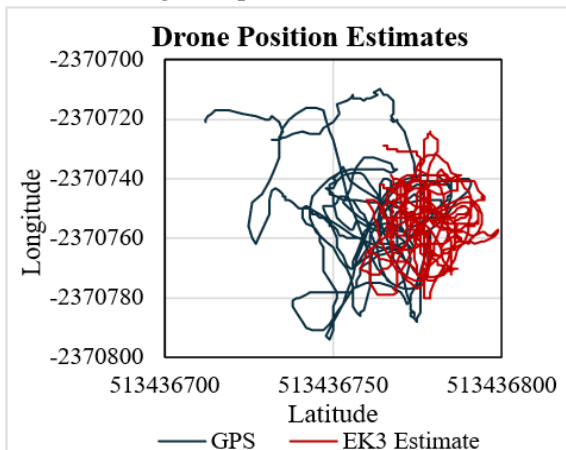


Figure 45: Plot displaying the drone position estimates.

5.3.3 Drone Stationary Vibrations

Vibrations during flight were recorded by the accelerometer in the X and Y direction, relative to the drone. The actuation of the RTI scanner was initiated at the midpoint of the loaded section shown.

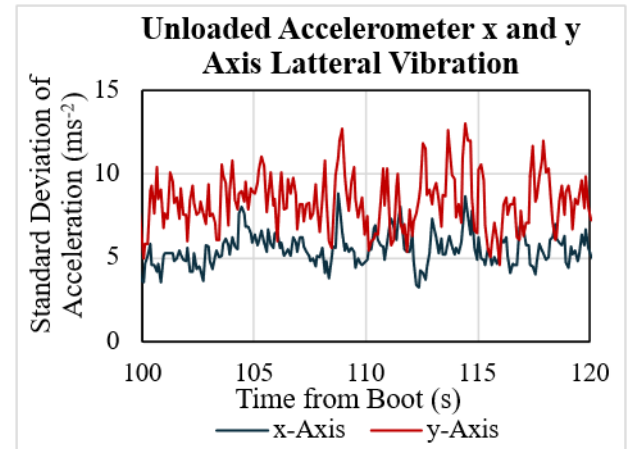


Figure 46: Graph displaying unloaded accelerometer x and y axes lateral vibrations.

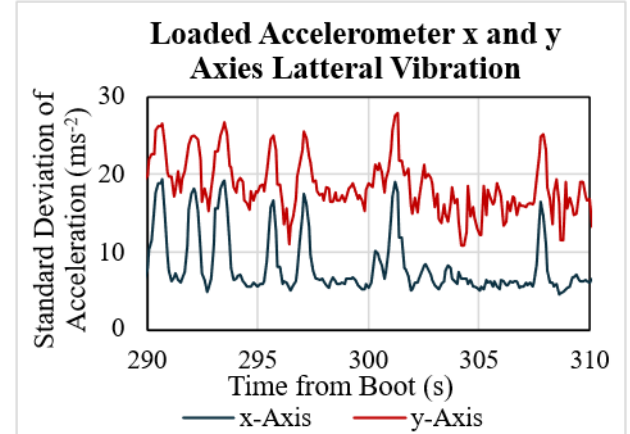


Figure 47: Graph displaying loaded accelerometer x and y axes lateral vibrations.

5.3.4 Stationary Optical Flow Performance

While hovering the optical flow sensors estimations of U and V velocities can be compared with those measured by the gyroscope to assess the stability of the drone in the X and Y directions in both cases.

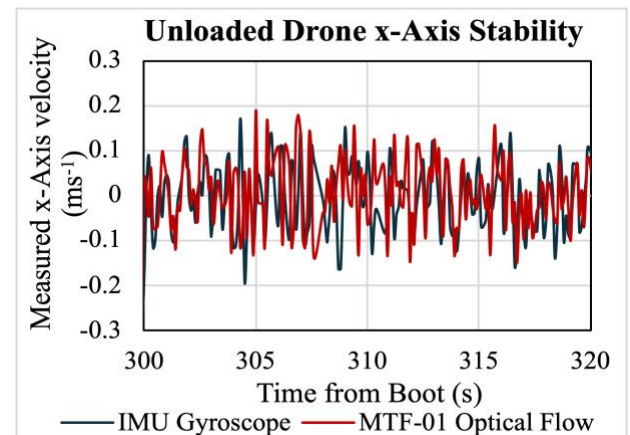


Figure 48: Graph showing unloaded drone x-axis stability.

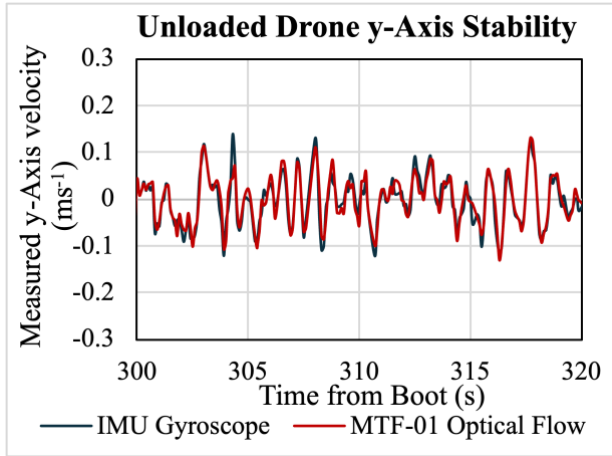


Figure 49: Graph showing unloaded drone y-axis stability.

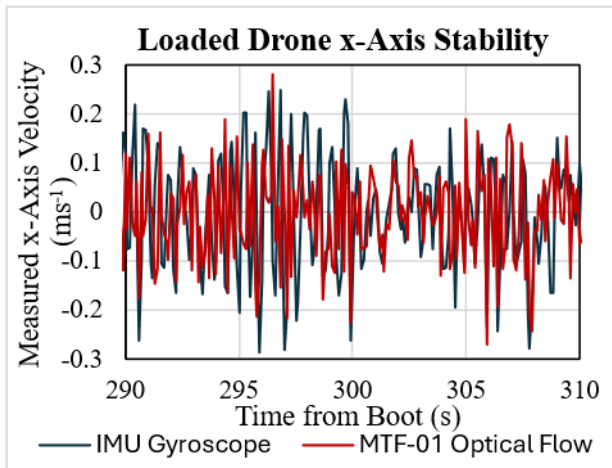


Figure 50: Graph showing loaded drone x-axis stability.

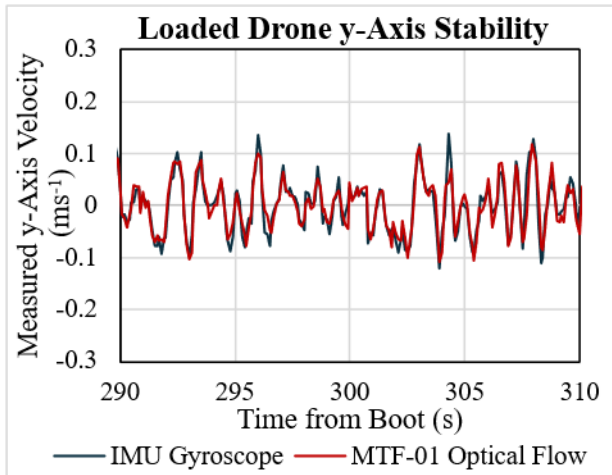


Figure 51: Graph showing loaded drone y-axis stability.

5.3.5 Autopilot Scripting Performance

As the purpose of the sensors is to be used with the RTI scanner, each test of the sensors ability of position hold was conducted during loaded testing. The front rangefinders distance in the X direction was recorded during the forward positioning autopilot:

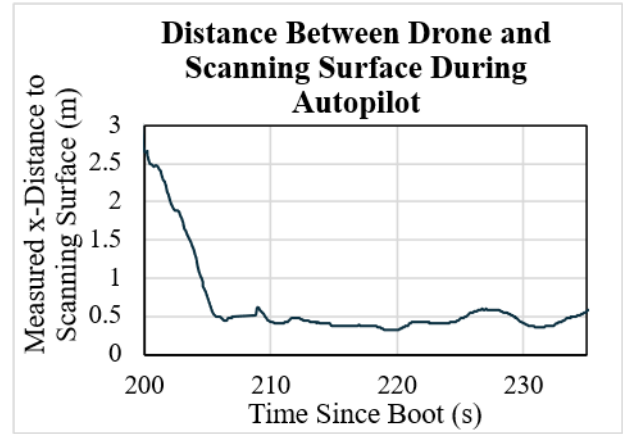


Figure 52: A graph displaying distance between drone and scanning surface during autopilot.

During which the EK3s positional estimate was also recorded:

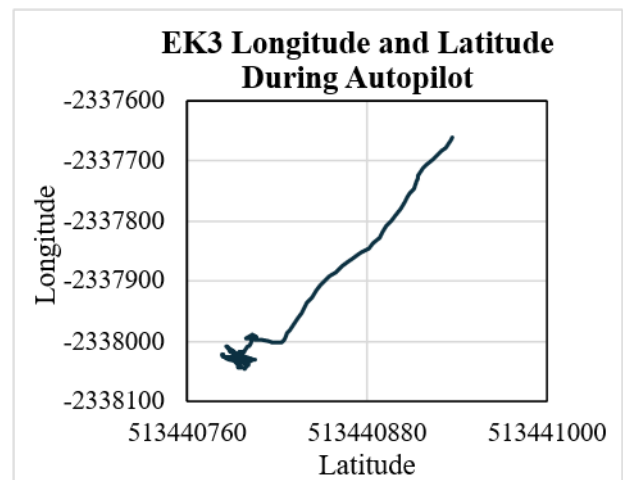


Figure 53: Plot displaying the EK3 longitude and latitude during autopilot.

6. DISCUSSION AND ANALYSIS

6.1 PRELIMINARY RTI TESTS

6.1.1 Distance Testing

RTI scan quality is highly dependent on the distance between the rangefinder and the scanned surface. Optimal results were obtained at distances between 300 mm and 350 mm.

The scan taken at 250 mm presented images that appear flatter due to shallow LED incident angles from the arm ends. This can lead to unusable light positions or may produce extensive shadows at edge surface protrusions, obscuring central features and decreasing the contrast of their shadows. Resulting MLIC and RTI output will provide lower quality results due to a lower quality and reduced data set.

Scans taken at 400mm also show unusable results. This can be attributed to the reduced angular illumination reducing the shadow diversity and

directional lighting. The decrease in light intensity with the inverse square law and the linear decrease of camera resolution with distance also massively decrease the quality of images processed.

Together, these adverse effects mean that shadows formed from illumination from a further distance are small and lack contrast, reducing the ability of the RBF algorithm to produce a re-lightable image from the MLIC which effectively shows surface features.

6.1.2 Ambient Light Intensity Testing

Optimal scanning is achieved in complete darkness as ambient light introduces multi-directional diffuse illumination, impairing the ability of the RTI software to simulate re-lighting from isolated vectors. Relatively low to moderate ambient light provides scans with an acceptable level of detail meaning that high quality results can be acquired outside after sunset, ideal for scanning wind turbine blades.

Scans taken with relatively high ambient light are not useable. This is due to diminishing shadow contrast, the introduction of non-representative shadows and disruptions to image processing from uncontrolled lighting vectors. Outdoor scans are therefore not appropriate in the daytime.

6.2 FINAL RTI TESTS

Figure 41 shows multiple surface scratches in the glossy polypropylene are revealed clearly only when dynamically re-lit. However, given the surface defects known on the sample used, this result represented a lower-than-expected improvement over visual inspection but did display a substantial improvement over a standard photograph. This surface scan fails to show variations in surface texture, but given the smoothness of the polypropylene, this is to be expected.

As observed from figure 42, the painted wall's texture is strongly enhanced using the proposed system, displaying bumps and paint deposits otherwise invisible in a simple photograph.

Comparing the images in figure 43, substantial surface roughness can be seen. Large surface textures visible to the eye are very apparent in the processed results and further smaller surface roughness not visible initially are clearly displayed after RTI processing.

It is therefore shown that the scanning provides very clear results on surface textures however does not display hairline scratches on a glossy surface clearly. Given that RTI typically provides the best resolution for fine details compared to other optical scanners [34], the lower-than-expected clarity of scratches can be attributed to a few sources. A substantial source of inaccuracy stems from the glossy surface of the polypropylene which reflects light rather than it diffusing across the scan surface. This leads to minimal shadows being cast across the area of interest and low contrast [58].

An element of inaccuracy could also stem from the camera used to capture the still frames. It not able to capture the recordings in raw video files, hence elements of camera processing and smoothing occur especially to compensate for low light exposure [59]. This means small shadows cast across the defects may have substantially less contrast in the processed images. Given images are gathered from frames of a recording, many of the images have elements of blur while the camera is adjusting, this will also reduce the shadow contrast in the images used.

Despite the lower-than-expected clarity, improvements can easily be made and the detection of small scratches on a smooth surface is established. Therefore, as a scanning method it would be appropriate for detecting defects expected on the surface of turbine blades.

6.3 DRONE PERFORMANCE ASSESMENT

The barometer and downward-facing rangefinder enable the drone to track and adjust its vertical altitude. The high and implausible variance of the barometer highlights its inherent inaccuracies, which result from environmental factors such as wind speed which affect surrounding air pressure. In contrast, the rangefinder provided values that more accurately represented the drone's motion and were less influenced by external conditions. The drone remained within an altitude range of 17cm for the duration of the results, however, would often maintain a near-stable position with only slight oscillations of 5cm observed. The frequency of these oscillations is too high to be plausibly attributed to the control system alone and is therefore likely caused by vibrations within the drone.

This 17 cm altitude variation is also prohibitively large for RTI scans, and the level of error is more likely rooted in limitations of the control system. When combined with the drone's mobility and the rangefinder's accuracy, this suggests inaccuracies within the EK3 estimate.

A comparison of the GPS and optical flow sensor performance clearly demonstrates the improved accuracy that results from optical inside-out tracking solutions, compared to GPS alone, as seen in the reduced spread of values. The enhanced tracking precision of the optical sensor also illustrates EK3's ability to fuse positional data from multiple sources, weighting them according to accuracy. However, drift is still present in the optical sensor, to a degree that is prohibitive for RTI. This is primarily observed immediately after take-off and can be attributed to limitations in the optical flow algorithm, which assumes consistent luminance of pixels between frames. This assumption is not a certainty and may fail under certain environmental conditions such as self-shadowing or reflective surfaces on the ground.

The oscillations observed in the altitude data can be partially explained by vibrations along the linear axes, both in loaded and unloaded configurations. According to ArduPilot's documentation [60], vibrational accelerations below 15 m/s^2 are ideal, with 30 m/s^2 being the maximum acceptable. While all measured vibrations were below 30 m/s^2 and often under 15 m/s^2 in optimal conditions, the precision required for a successful RTI scan means that even these vibration-induced errors within the EK3 are a significant limiting factor in achieving the necessary accuracy.

Notably, vibrations were greater along the X-axis. This is likely due to a more concentrated mass distribution in that direction, increasing the moment of inertia. While a higher moment of inertia typically dampens vibrations, it also impairs the drone's ability to turn, as demonstrated in the stability plots.

The optical flow readings further suggest that uneven mass distribution affects stability, with Y-axis stability consistently exceeding that of the X-axis. Moreover, the difference in X-axis stability between the loaded and unloaded cases is greater than the corresponding difference in Y-axis stability, implying that adding mass along the X-axis has a more substantial impact. The optical flow sensor also proves more useful than the IMU alone, as its high

polling rate offers a more accurate representation of X-axis stability.

Finally, the front-facing rangefinder demonstrated high accuracy, providing precise distance measurements that allowed the drone to avoid potential collisions. These measurements were processed onboard by the flight controller with negligible latency. However, due to the increased precision requirements of RTI, particularly due to the sensitivity of the distance between the camera sensor and scanning surface, a more reliable system would be needed to produce readable scans. Although the drone achieved sub-20 cm accuracy in the X-direction, this is insufficient. As shown in the latitude and longitude plots, this accuracy also came at the expense of motion precision along the Y-axis.

Lastly, the premature activation of the low voltage failsafe implies that the 6s battery used was unable to supply its rated voltage for a prolonged period under high throttle conditions before the voltage lowered precipitously due to battery sag. Leaving up to 40% of its capacity unused. Thus, limiting the amount of time available for a potential scan

6.4 LIMITATIONS AND IMPROVEMENTS

6.4.1 Scanner Improvements

Testing the constructed scanner provided accurate results, however also exposed several limitations in this element of the prototype. These can largely be attributed to limited availability of component selection and access to manufacturing methods, which are both related to the imposed budget and time constraints. The following issues have been identified, with recommendations provided to address these.

This first issue discovered from testing was the components' wear over time, with deformations observed in 3D printed parts comprising the rotating arm mechanism, causing increased friction, dislodgement of components and, at times, fractures, all of which were detrimental to performance. This issue was particularly noticeable with the belt gears and the surrounding supporting components. One reason for this is the use of PLA, which was found to be too brittle for mechanical components like gears. For a final design, alternative polymers/ polymer composites can be used. The most suitable of these is carbon reinforced polyamide (nylon) which has improved toughness and reduced friction from

natural lubrication [61]. The RTI arm attachment to the timing belt was also identified as another source of failure, with the load bearing points being susceptible to shear fractures. Alternative material choices could again include carbon reinforced nylon, or high strength polymers like polycarbonate (PC) or polyether ether ketone (PEEK) [62, 63].

Structural components like the tracks or support frames were found to be adequate for their purpose after minimal repeated use, however regular extended use would likely result in all components failing at a given point. Improvements on the outlined construction should incorporate materials of higher toughness, particularly in load-bearing points on the mechanism, mitigating the formation of shear fractures. The aforementioned materials would be suitable alternatives. To reduce the friction of the belt as it passes around the track, the contact surface should be lined with a low friction material like aluminium [63].

The primary mode of component failure while testing was due to sudden increases shear forces. Layering from 3D printing was found to be the reason for this, as the strength between the layers isn't as high as the material itself. On each occasion a location of failure was found at a point of change of geometry where a separation of layers could be seen. Bending can also be caused by the separation of print layers. This was commonly displayed with the slumping of the RTI arms as the track connector bent, leading to inconsistent light positions. Alternative manufacturing methods like injection moulding or milling should be used in a final design depending on the material selected and component of interest.

Alternate manufacturing methods will also reduce the tolerance of the parts produced [64]. The performance of 3D printed parts varied substantially based on the given dimensions of a print. This most appeared in the connectors of the arm and the track and caused some wheels to lift and not lay flush against the track, resulting in instabilities and vibrations, impacting the quality of scan results.

In testing, many wires had to be accounted for, and a large Arduino and circuit board had to be accommodated. This massively increases the risk of cables being cut or the boards interfering. Using a

designed printed circuit board would allow all the components to be included on a compact board. A more compact board would also be compatible with a smaller microcontroller like an Arduino Pico. Boards this size could be housed within the pre-existing drone body, reducing its interference and allowing for better wire management.

During testing it could also be seen that the camera shifted during stationary operation. This is caused by the LED cables pushing on the camera and therefore, this also poses a risk to position hold interference during flight tests. To overcome this issue, RTI arms should be used that house the LED cables and move them away from interfering with key components and sensors.

An issue found while testing and processing results was the use of an open loop controlled RTI arm actuation. This led to inconstant arm positioning, regular calibrations, components being under excess force and reductions in performance when the battery is not at full charge. Feedback sensors should be added to make a closed control loop to improve accuracy. Methods for feedback could include sensors at desired positions or to include a sensor by the gears to count the rotations. Position sensors would ensure positional consistency while, counting the rotations of the gears would allow the selection of as many positions as desired. In either case detectors should be included at the top and bottom to ensure a full range of motion and a return to the start position in each cycle. As shown previously the constancy of lighting positions will mean no calibration would be needed in the case of unexpected arm shifts.

6.4.2 Drone Improvements

The drone performance results validate the use of an optical sensor to improve positional tracking—from meter-level resolution to position-hold performance within 20 cm. However, this level of precision remains insufficient for obtaining RTI scans in its current form. This limitation is primarily due to vibrations and uneven mass distribution induced by mounting a scanning payload on the drone as well as the limitations of the specific sensors chosen.

Improvements can be made by reducing structural vibrations that are detected by sensors and mistakenly incorporated into the EK3 calculations.

One such improvement is the use of soft rubber mounts to isolate sensitive sensors, such as rangefinders, thereby providing shock absorption.

Since the optical flow sensor exhibited drift in its measurements, a more bespoke, higher-resolution assembly could be considered.

Furthermore, because the design centres its mass along the X-axis, this information could be incorporated into PID tuning to enhance the stabilization performance of the optical sensor along that axis.

Alternatively, a larger drone platform—such as a 7-inch frame—could be used to scale the design. This would provide additional space to more evenly distribute the mass of the scanning components.

Finally, the use of high discharge polymer batteries with low internal resistances would mitigate the battery sag currently experienced, enabling longer flight times by utilising more of the battery's rated capacity. This would come at the expense of a heavier battery and higher overall weight. However, the current relatively high loaded thrust-to-mass ratio confirm that this is a viable solution.

6.5 FUTURE WORKS

The prototype developed lays the groundwork for the unique concept of incorporating drone-mounted RTI scanner setups to assess early signs of damage on structures otherwise difficult to reach. With further design iterations and developments, this could build into a drone-mounted RTI scanner used in industry and have applications far beyond our specific intended use case. Future works in this field to push progress forward first includes improving stability to a degree which permits successful scanning, even under expected high wind speed conditions experienced in the intended use case.

The initial work that must be done is to ensure a reliable control system and structural stability for the RTI arm actuation, so the lights are always in the same positions relative to the camera. Consistency removes the need for regular calibration and will allow for complete freedom to test in a range of situations beyond static testing and basic flight tests.

The largest hurdle to overcome is the stability of the drone, which is currently limited by the performance of available inside-out tracking solutions. Enterprise systems, such as Simultaneous Localization and Mapping (SLAM), build upon optical flow by incorporating multiple cameras to construct a 3D map of the surrounding environment, allowing for more precise orientation and positioning calculations [65].

The results obtained using optical flow demonstrate the ability of modern flight controller microprocessors, such as the STM32H7, to compute positional information from a variety of sources. Work by other authors highlights the potential of neural network imaging techniques to enhance image-based tracking and positional accuracy [66].

Once these parameters are met scans can be conducted on either turbine blades or a comparable material while the drone is in flight to directly solve problems mentioned previously in the report. Once constructed and scans are verified further additions to the scanner should be made. These include adapting the scanner to capture a wide range of angles to accommodate the contours of a wind turbine blade or the use of machine learning for automated crack detection. The design could also be adapted for uses in areas beyond wind turbine blades, especially in circumstances where damage detection is needed in areas of difficult to reach geometry like buildings or bridges.

7. CONCLUSION

This project set out to develop a quadcopter-mounted Reflective Transformation Imaging (RTI) scanner for identifying early-stage surface damage on wind turbine blades. By combining Unmanned Aerial Vehicle (UAV) technology with this computational imaging technique, a proof-of-concept prototype was designed, tested, and evaluated under controlled conditions. The system successfully generated Multi-Light Image Collections (MLICs) using the assembled prototype, which were processed with Relight software, revealing surface defects and details otherwise imperceptible using standard photography. These results demonstrate the potential of RTI scanning for the intended application, offering an innovative approach to non-destructive testing (NDT) in structural health monitoring of wind turbine blades.

Despite its success in static testing, the prototype encountered flight stability challenges that prevented reliable scanning in dynamic conditions. Issues such as vibrations, optical tracking drift, and mass distribution imbalance interfered with position hold functionality, leading to inconsistent scanning distances during flight. Additionally, early battery failsafe activation significantly limited operational time, restricting the system's ability to successfully perform full-scale RTI scans. These findings highlight the need for refinement in flight control and structural improvements to optimise the drone's stability, enabling precise imaging.

Key Findings:

1. Prototype validation: The developed system successfully produced RTI-enhanced images across a variety of surface types, demonstrating its ability to highlight subtle surface variations.
2. Flight stability limitations: Despite achieving sub-20cm accuracy, the inside-out tracking system exhibited positional drift, preventing the drone from maintaining a fixed scanning distance.
3. Mechanical challenges: The RTI arm mechanism enabled adaptable lighting control, but exhibited structural weaknesses after repeat testing, indicating material and manufacturing process optimisation is required to reduce wear, prevent deformation, and improve durability for consistent scanning performance.
4. Battery and power constraints: The drone's high-current draw under payload conditions triggered early voltage failsafe activation, reducing flight duration.

Proposed Improvements:

1. Advanced tracking and stability enhancements: Implementing experimental inside-out tracking solutions such as Simultaneous Localisation and Mapping (SLAM) tracking will improve positional accuracy and reduce drift issues.
2. Material and Structural Optimization: Replacing PLA components with carbon-reinforced nylon or PEEK will increase mechanical durability and reduce deformation risk.
3. Closed-loop arm control: Introducing feedback-based servo actuation will ensure consistent light positioning, eliminating variability stemming from the open-loop system.
4. Aerodynamic refinements: Adjusting mass distribution and reinforcing structural supports will mitigate oscillations, ensuring precise scanning distances.
5. Battery optimization: Using high-discharge lithium-polymer batteries will extend flight duration, enabling complete scans to be performed without premature landing.

This research sets the foundation for integrating UAVs and RTI scanning, demonstrating potential for automated detection of surface damages in wind turbine blades. Future work should expand the scope of this project, attempting tests on real wind turbine blades at high altitudes in the environments they're installed, where higher-speed turbulent winds are present. Additionally, refining the RTI system's scan automation, combined with the use of machine learning-based defect classification, could enable fully autonomous inspections. The drone-mounted RTI scanner could also be adapted for broader applications, such as civil infrastructure and offshore installations, further improving its practical utility.

8.AUTHOR CONTRIBUTIONS

All authors contributed to the work presented, including research, manufacture, testing and report writing. A breakdown of the primary responsibilities is provided:

Thomas Brown - Research and development of the RTI scanner

Niall Fatah - Research and development of the drone

Hasan Hachem – Background research into the use of drones and NDT methods, support in the development of the final prototype.

Harvey Chadderton – Background research into UAVs and NDE methods, support in developing and testing

Finn Hadley-Burgess - Research and usage of RTI software for MLIC processing

9.DECLARATIONS

All authors declared no conflict of interest.

10.ACKNOWLEDGMENTS

The authors acknowledge and express gratitude to the supervision provided by Associate Professor Adam Wojcik for his guidance and support in undertaking this project. Acknowledgment is paid to University College London's broader Department of Mechanical Engineering and its staff, particularly those at the Materials Characterisation Lab, for providing initial workbench space. A stoichiometric ratio of caffeine, music and good humour are credited for the authors' perseverance through both this project and the wider master's degree it marks the end of. Finally, the contribution of each group member's family and friends is recognised for their patience and support.

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12. APPENDIX

RTI Arduino Code

```
#include <Servo.h>
```

```
const int dataPin = 11;
```

```
const int clockPin = 12;
```

```
const int latchPin = 8;
```

```
const int servoPin = 9;
```

```
Servo myServo;
```

```
void setup() {
```

```
  pinMode(dataPin, OUTPUT);
```

```
  pinMode(clockPin, OUTPUT);
```

```
  pinMode(latchPin, OUTPUT);
```

```
  myServo.attach(servoPin);
```

```
  shiftOutData(0x00);
```

```
}
```

```
void loop() {
```

```
  for (int i = 0; i < 6; i++) {
```

```
    myServo.write(180); // Servo speed
```

```
    delay(1200); // Time in motion for each step
```

```
    myServo.write(90);
```

```
    delay(100);
```

```
    // Cycle the LEDs after each step
```

```
    for (byte i = 0; i < 8; i++) { // Cycle first 8 LEDs
```

```
      byte data = (1 << i);
```

```
      shiftOutData(data);
```

```
      delay(3000); // Time of LED blink
```

```
      shiftOutData(0x00);
```

```
      delay(5);
```

```
    }
```

```
    for (byte i = 8; i < 16; i++) { // Cycle second 8 LEDs
```

```
      byte data = (1 << (i - 8));
```

```
      shiftOutData(data << 8);
```

```
      delay(3000); // Time of LED blink
```

```
      shiftOutData(0x00);
```

```
      delay(5);
```

```
    }
```

```
  }
```

```
  //Servo return to original position
```

```
  for (int i = 0; i < 1; i++) {
```

```
    myServo.write(0); // Servo speed
```

```
    delay(7200); //Time servo moves for
```

```
    myServo.write(90);
```

```
    delay(100);
```

```
  }
```

```
  // LED cycle at initial position
```

```
  myServo.write(90);
```

```
  // Cycle through the first 8 LEDs
```

```
  for (byte i = 0; i < 8; i++) {
```

```
    byte data = (1 << i);
```

```
    shiftOutData(data);
```

```
    delay(500); // Time of LED blink
```

```
    shiftOutData(0x00);
```

```
    delay(5);
```

```
  }
```

```
  // Cycle through the second 8 LEDs
```

```
  for (byte i = 8; i < 16; i++) {
```

```
    byte data = (1 << (i - 8));
```

```
    shiftOutData(data << 8);
```

```
    delay(500); // Time of LED blink
```

```
    shiftOutData(0x00);
```

```
    delay(5); // Wait
```

```
  }
```

```
  delay(1000);
```

```
  // All LEDs blink to show completion
```

```
  for (int i = 0; i < 3; i++) {
```

```

    shiftOutData(0xFFFF);
    delay(500); // Time of LED blink
    shiftOutData(0x0000);
    delay(100);
}

// Stop execution after full cycle
while (true) {

}

}

void shiftOutData(uint16_t data) {
    digitalWrite(latchPin, LOW);
    shiftOut(dataPin, clockPin, MSBFIRST,
highByte(data));
    shiftOut(dataPin, clockPin, MSBFIRST, lowByte(data));
    digitalWrite(latchPin, HIGH);
    delayMicroseconds(10);
}

```

Drone Custom x-Axis, Scanning Object Proximity Script

```

-- Loiter Mode Wall Distance Hold
Script
-- This script keeps the copter at
approximately 50cm from a wall when in
Loiter mode and RC Channel 7 is
enabled.
-- It uses Rangefinder 2 (forward-
facing) for distance measurement and
overrides pitch control to maintain the
set distance.

-- Constants and configuration
local LOITER_MODE = 5 -- ArduCopter
Loiter mode
number:contentReference[oaicite:2]{inde
x=2}
local RANGEFINDER_ORIENT = 0 --
Orientation code for forward-facing
rangefinder (0 =

```

```

forward):contentReference[oaicite:3]{in
dex=3}
local TARGET_DISTANCE_CM = 100 --
Desired distance from wall in
centimeters
local ACTIVATION_CHANNEL = 7 -- RC
channel used to activate this script
(e.g., aux switch on channel 7)
local PWM_CENTER = 1500 --
Neutral PWM value (no pitch input)
local PWM_MIN = 1400 --
Minimum PWM for backward movement
(limited for safety)
local PWM_MAX = 1600 --
Maximum PWM for forward movement
(limited for safety)
local DEADZONE_CM = 5 --
Tolerance in cm around target distance
to avoid oscillation

-- Retrieve RC channel object for pitch
control (channel 2 is typically pitch)
local rc_pitch = rc:get_channel(2) --
RC Channel 2 (Pitch) object for
override control

-- Main update function (periodically
called by the scheduler)
function update()
    -- Only operate when the vehicle is
armed and in Loiter mode
    if not arming:is_armed() or
vehicle:get_mode() ~= LOITER_MODE then
        -- Ensure no override if not in
proper state
        rc_pitch:set_override(0) -- 0
(or nil) will clear any override on the
pitch
channel:contentReference[oaicite:4]{ind
ex=4}
        return update, 100 -- Run at
10Hz until conditions are met
    end

    -- Vehicle is in Loiter mode and
armed
    local ch7_pwm =
rc:get_pwm(ACTIVATION_CHANNEL)

```

```

    if ch7_pwm and ch7_pwm > 1700 then
-- Activation switch is ON (high
position)
        -- Read the forward distance
from rangefinder (in cm)
        local dist_cm =
rangefinder:distance_cm_orient(RANGEFIN
DER_ORIENT) -- Distance to obstacle in
front:contentReference[oaicite:5]{index
=5}
        if dist_cm and dist_cm > 0 then
            -- Calculate error between
current distance and target distance
            local error = dist_cm -
TARGET_DISTANCE_CM
            if math.abs(error) <=
DEADZONE_CM then
                -- Within tolerance -
hold position (no pitch movement)

rc_pitch:set_override(PWM_CENTER) --
Keep pitch stick centered (no movement
override)
            else
                -- Determine pitch
override direction and magnitude
                -- Positive error (too
far): pitch forward (increase PWM above
center)

                -- Negative error (too
close): pitch backward (decrease PWM
below center)
                local correction =
error -- Proportional control: 1 PWM
unit per 1 cm error (tunable)
                local target_pwm =
PWM_CENTER + correction

                -- Constrain the PWM to
safe limits to avoid aggressive moves
                if target_pwm > PWM_MAX
then
                    target_pwm =
PWM_MAX
                elseif target_pwm <
PWM_MIN then
                    target_pwm =
PWM_MIN
                end
            end
        end
    end

```

```

rc_pitch:set_override(target_pwm) --
Override pitch channel to adjust
distance:contentReference[oaicite:6]{in
dex=6}
        end
    else
        -- No valid rangefinder
data (sensor out of range or no
reading)

        -- Release pitch override
to give pilot full control in this case
        rc_pitch:set_override(0) -
- Clear override if distance is
invalid/unavailable
    end
else
    -- Activation switch is OFF
(Channel 7 low) -> do not override,
allow normal control
    rc_pitch:set_override(0) --
Ensure pitch override is cleared when
not active
end

    return update, 50 -- Schedule next
update in 50ms (20Hz loop for
responsive control)
end

return update, 50 -- Initial scheduler
call at 20Hz

```